MA204 Abstract Algebra

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Notation

$\mathbb{R}$	the set of real numbers,
$\mathbb{R}^*$	the set of non-zero real numbers,
$\mathbb{R}^+$	the set of positive real numbers,
$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$	the set of complex numbers,
$\mathbb{C}^*$	the set of non-zero complex numbers,
$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$	the set of integers,
$\mathbb{Z}^{\geq 0} = \{0, 1, 2, 3, \ldots\}$	the set of non-negative integers,
$\mathbb{N} = \mathbb{Z}^+ = \{1, 2, 3, \ldots\}$	the set of natural numbers (= the set of positive integers),
$\mathbb{Q} = \{a/b \mid a, b \in \mathbb{Z}, b \neq 0\}$	the set of rational numbers,
$\mathbb{Q}^+ = \{a/b \mid a, b \in \mathbb{Z}, b \neq 0, a/b > 0\}$	the set of positive rational numbers,
$M_{m\times n}(R)$	the set of $m \times n$ matrices (ie m rows, n columns) with entries from R, where $R = \mathbb{Z}, \mathbb{Q}, \mathbb{C}$, etc.
$n\mathbb{Z} = \{\ldots, -3n, -2n, -n, 0, n, 2n, 3n, \ldots\}$	the set of integer multiples of the integer n

$$a^m = \begin{cases} \underbrace{a * \cdots * a}_{m} & m > 0 \\ \text{identity} & m = 0 \\ \underbrace{a^{-1} * \cdots * a^{-1}}_{-m} & m < 0 \end{cases}$$

Section 1. Introduction.

Many mathematical objects and operations that you have studied so far (matrices, real numbers, integers, vectors, with addition, multiplication) and many others that you have not yet encountered can be seen to obey a common set of rules, and behave in similar ways. Abstract algebra is a branch of mathematics that provides a general framework for studying such mathematical systems. The idea is that we shouldn't be developing a mathematical theory over and over again for the different systems we encounter, when really they are all special cases of a more general system. Abstract algebra is about 'abstracting' the rules of other systems, developing a theory of them that can be applied to the original systems. So 'abstract' should be seen as a good thing, as it can be applied in many contexts in a very 'concrete' way.

Section 2. Binary Operations.

Definition 2.1 (Closed binary operation)

A *closed binary operation* on a set S is a rule that assigns to each ordered pair (a, b) (where $a, b \in S$) some element of S.

(We sometimes instead say that a binary operation is *closed on a set* S.)

Example 2.2 (Examples of closed binary operations)

(a) Let $S = \mathbb{R}$, the set of real numbers. The addition operation '$+$' is a closed binary operation on S. The same is true if we replace $\mathbb{R}$ by any of $\mathbb{C}$ (the set of complex numbers), $\mathbb{Z}$ (the set of integers), $\mathbb{N} = \{1, 2, 3, \ldots\}$ (the set of natural numbers).

(b) Let $S = \mathbb{R}$, the set of real numbers. The subtraction operation '$-$' is a closed binary operation on S. The same is true if we replace $\mathbb{R}$ by any of $\mathbb{C}$ (the set of complex numbers), $\mathbb{Z}$ (the set of integers). However, it is not true if we replace $\mathbb{R}$ by $\mathbb{N}$ since, for example $2 - 3 \notin \mathbb{N}$.

(c) Let $S = \mathbb{R}^*$, the set of non-zero real numbers. Then '$+$' is not a closed binary operation on S because (for example) $3 + (-3) = 0 \notin S$.

(d) Let $S = M_{2\times 2}(\mathbb{R})$ be the set of all 2×2 matrices with real entries. Then the operation of matrix addition is a closed binary operation on S. However, if we let $S = M(\mathbb{R})$ be the set of all matrices with real entries, then the operation of matrix addition is not a closed binary operation on S, since $A + B$ is not defined for matrices A, B of different sizes.

(e) The operation of taking the absolute value of a real number is not a binary operation (at all), as it takes *one* real number and produces a real number.

(f) The operation of taking the determinant of a matrix (with real entries, say) is not a binary operation (at all), as it takes *one* matrix and produces a real number.

Example (Example 2.2 continued)

(g) Let F denote the set of all real-valued functions f whose domain is $\mathbb{R}$. For each $f, g \in F$, and for each $x \in \mathbb{R}$ define

$$\begin{aligned}(f+g)(x) &= f(x)+g(x) \quad \text{function addition,}\\ (f-g)(x) &= f(x)-g(x) \quad \text{function subtraction,}\\ (f\cdot g)(x) &= f(x)\cdot g(x) \quad \text{function multiplication,}\\ (f\circ g)(x) &= f(g(x)) \quad \text{function composition.}\end{aligned}$$

Then for $f, g \in F$ we have that $f+g, f-g, f\cdot g, f\circ g$ are all real valued functions with domain $\mathbb{R}$, so are elements of F. Thus these operations $+, -, \cdot, \circ$ are closed binary operations on F.

(h) Let $S = \mathbb{Z}$ and define the operation $*$ by setting $a * b$ to be the smaller of a or b, or either if $a = b$. (That is, we set $a * b = \min\{a, b\}$). Then '$*$' is a closed binary operation on $\mathbb{Z}$.

(i) Let $S = \mathbb{Z}$ and define the operation $*$ by setting $a * b = a$. Then '$*$' is a closed binary operation on $\mathbb{Z}$.

Definition 2.3 (Commutative Operation)

A binary operation $*$ on a set S is *commutative* if $a * b = b * a$ for all $a, b \in S$.

If $a * b = b * a$ then we say that a and b commute, or that a *commutes with* b (and that b *commutes with* a).

Example 2.4 (Commutative Operation)

(a) Let $S = \mathbb{Z}$, then $a + b = b + a$ for all $a, b \in S$, so '$+$' is commutative. However $a - b \neq b - a$ for all $a, b \in S$ so '$-$' is not commutative.

(b) Let $S = M_{n\times n}(\mathbb{R})$ (the set of all $n \times n$ matrices with real entries). Then $A + B = B + A$ for all $A, B \in S$ so matrix addition is commutative; however $A \cdot B \neq B \cdot A$ for all $A, B \in S$, so matrix multiplication is not commutative.

(c) Let $S = \mathbb{Z}$ and define the operation $*$ by setting $a * b = a$. Then if $a \neq b$ we have $a * b \neq b * a$ so $*$ is not commutative.

Definition 2.5 (Associative Operation)

A binary operation $*$ on a set S is *associative* if for all $a, b, c \in S$ we have $(a * b) * c = a * (b * c)$.

Example 2.6

(a) Addition of real numbers is associative: $(a + b) + c = a + (b + c)$. However, subtraction of real numbers is not associative: $(a - b) - c \neq a - (b - c)$, in general.

(b) It follows from (a) that addition of matrices (of the same size) with real entries is associative. Moreover, multiplication of $n \times n$ matrices is associative. For 2×2 matrices this is routine to check, i.e. to verify that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \left(\begin{pmatrix} e & f \\ g & h \end{pmatrix} \cdot \begin{pmatrix} i & j \\ k & l \end{pmatrix} \right) = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & h \end{pmatrix} \right) \cdot \begin{pmatrix} i & j \\ k & l \end{pmatrix}$$

Throughout this module you may state without proof that matrix multiplication (where defined) is associative.

Example (Example 2.6 continued)

(c) (Function composition is associative). We have

$$f \circ (g \circ h) = (f \circ g) \circ h$$

whenever this composition is defined. We can see this as follows. Let $\theta = f \circ g$, $\phi = g \circ h$. Then

$$(f \circ (g \circ h))(x) = (f \circ \phi)(x) = f(\phi(x)) = f((g \circ h)(x)) = f(g(h(x)));$$
$$((f \circ g) \circ h)(x) = (\theta \circ h)(x) = \theta(h(x)) = (f \circ g)(h(x)) = f(g(h(x))).$$

Thus $(f \circ (g \circ h))(x) = ((f \circ g) \circ h)(x)$ for all x so

$$(f \circ (g \circ h)) = ((f \circ g) \circ h).$$

We can define closed binary operations on a set S in terms of multiplication tables where the rows and columns are indexed by the elements of S and the entries are elements of S. We shall use the convention that for $x, y \in S$, $x * y$ is the element in intersection of the row indexed by x and the column indexed by y.

Example 2.7

(a) Consider the table below:

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

Then, for example, $a * b = c$, $c * b = b$. Note that $b * a \neq a * b$ so the operation $*$ is not commutative and that $(a * a) * a = b * a = a$ and $a * (a * a) = a * b = c$ so $*$ is not associative.

(b) Consider the table below:

$*$	a	b	c	d	e
a	a	b	c	b	d
b	b	c	a	e	c
c	c	a	b	b	a
d	b	e	b	e	d
e	d	c	a	d	c

Then $*$ is commutative (since the table is symmetric across the leading diagonal). It is not associative since, for example, $(b * d) * c = e * c = a$ but $b * (d * c) = b * b = c$.

Exercise 2.8

In each of the cases below determine if the binary operation is commutative and if it is associative:

(a) $*$ defined on $\mathbb{Q}$ (the set of rational numbers a/b, $a, b \in \mathbb{Z}$) given by $a * b = ab + 1$;

(b) $*$ defined on $\mathbb{Q}$ given by $a * b = ab/2$;

(c) $*$ defined on $\mathbb{N}$ given by $a * b = 2^{ab}$;

(d) $*$ defined on $\mathbb{N}$ given by $a * b = b^a$.

Exercise 2.9

(a) Let $S = \{a, b\}$. How many closed binary operations on S are there?

(b) Let $S = \{a, b, c\}$. How many closed binary operations on S are there?

(c) Let S be a set of n elements. How many closed binary operations on S are there?

Exercise 2.10

Complete the table below so that it defines a commutative binary operation on $S = \{a, b, c, d\}$.

$*$	a	b	c	d
a	a	b	c	
b	b	d		c
c	c	a	d	b
d	d			a

Exercise 2.11

Let A be a finite set and let $a \in A$ be some fixed element of A. Let $*$ be a binary operation defined by $x * y = a$ for all $x, y \in A$. Show that $*$ is associative.

Exercise 2.12

Light's associativity test is the following. Let $\cdot$ be a binary operation defined on a finite set A. For each $a \in A$ define an operation $*_a$ by $x *_a y = x \cdot (a \cdot y)$ and define an operation $\circ_a$ by $x \circ_a y = (x \cdot a) \cdot y$. If there exist $a, x, y \in A$ such that $x *_a y \neq x \circ_a y$ then $x \cdot (a \cdot y) \neq (x \cdot a) \cdot y$ so $\cdot$ is not associative. On the other hand, if $x *_a y = x \circ_a y$ for all $x, y, a \in A$ then $x \cdot (a \cdot y) = (x \cdot a) \cdot y$ for all $x, y, a \in A$ so $\cdot$ is associative. It is possible to determine if these conditions hold by finding the multiplication tables of $\circ_a$ and $*_a$ for all $a \in A$.
Let $A = \{a, b\}$ and define the binary operation $\cdot$ on A by the multiplication table below.

$\cdot$	a	b
a	a	b
b	b	a

Use Light's associativity test to determine whether or not $\cdot$ is associative.

Section 3. Groups.

The first type of algebraic object we will discuss are called groups, and these are systems with a single operation. (Later in the module we will discuss objects, called rings, with two operations.)

Definition 3.1 (Group)

A *group* is a set G together with a binary operation $*$ on G such that the following hold:

(i) G is *closed* under $*$ (i.e. for all $g, h \in G$ we have $g * h \in G$);

(ii) $*$ is associative;

(iii) there is an element $e \in G$ such that for all $g \in G$ we have $e * g = g * e = g$ (such an element e is called an *identity* of G);

(iv) for each $g \in G$ there is an element $g' \in G$ such that $g * g' = g' * g = e$. (The element g' is called the *inverse of* g with respect to the operation $*$.)

Often we just talk about a group G, if it is clear what the operation concerned is, and when it is clear what the operation $*$ is, we will often write gh instead of $g * h$, for example.

Definition 3.2 (Abelian group)

A group is *abelian* if its binary operation is commutative.

Example 3.3

(a) The set of integers $\mathbb{Z}$ with operation $+$ is a group: (i) $\mathbb{Z}$ is closed under addition, (ii) $+$ is associative, (iii) the identity element is $0 \in \mathbb{Z}$, (iv) the inverse of an element $a \in \mathbb{Z}$ is $-a \in \mathbb{Z}$.

(b) The natural numbers $\mathbb{N}$ with operation $+$ is not a group as it has no identity element.

(c) The non-negative integers with operation $+$ is not a group: its identity is 0, but the absence of negative numbers means that no positive number has an inverse.

(d) The set of integers $\mathbb{Z}$ with operation $\cdot$ (multiplication of integers) is not a group: the identity element would be 1, but the element $2 \in \mathbb{Z}$ (for example) has no inverse. (In fact, only 1 and -1 have multiplicative inverses.)

(e) The set of non-zero real numbers $\mathbb{R}^*$ with operation $\cdot$ (multiplication) form a group: (i) multiplying two non-zero real numbers gives a non-zero real number, (ii) multiplication of real numbers is associative, (iii) the non-zero real number 1 is the identity, (iv) for an non-zero real number a, its multiplicative inverse is $1/a$, which is a non-zero real number.

(f) The set $M_{m\times n}(\mathbb{R})$ of all $m \times n$ matrices with real entries with the operation of matrix addition forms a group: (i) the sum of two matrices in $M_{m\times n}(\mathbb{R})$ is a matrix in $M_{m\times n}(\mathbb{R})$, (ii) addition of matrices is associative, (iii) the identity is the zero matrix (ie the matrix all of whose entries is zero), (iv) the inverse of $A \in M_{m\times n}(\mathbb{R})$ is $-A \in M_{m\times n}(\mathbb{R})$.

Example (Example 3.3 continued)

(g) The set $M_{n\times n}(\mathbb{R})$ of all $n \times n$ (square) matrices with real entries with the operation of matrix multiplication does not form a group since (for example), the identity element is the $n \times n$ identity matrix but the matrix with all entries equal to zero does not have an inverse.

(h) The set of complex numbers $S = \{1, -1, i, -i\}$ with the operation of multiplication of complex numbers forms a group: (i) multiplying any two elements of S gives another element of S, (ii) multiplication of complex numbers is associative, (iii) the number $1 \in S$ is the identity, (iv) the inverse of 1 is 1, the inverse of -1 is -1, the inverse of i is $-i$, the inverse of $-i$ is i.

(i) The set of complex numbers $S = \{e^{i\theta} \mid 0 \leq \theta < 2\pi\}$ with the multiplication operation forms a group. Parts (i),(ii),(iii) are checked as in (h), for (iv) the inverse for $e^{i\theta}$ is $e^{i(2\pi-\theta)}$ for $\theta \neq 0$, and the inverse of 1 is 1.

(j) The set of matrices $S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ under matrix multiplication forms a group: (i) multiplying any two elements of S gives another element of S, (ii) multiplication of matrices is associative, (iii) the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in S$ is the identity, (iv) the inverse of $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the inverse of $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ is $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

Example 3.4

We show that the set S of all non-singular $n \times n$ matrices with real entries forms a group under matrix multiplication. (Recall that a matrix is *non-singular* if its determinant is non-zero.)

(i) To prove that S is closed under matrix multiplication we must show that if A, B are non-singular then AB is non-singular, but this follows since $\det(AB) = \det(A)\det(B) \neq 0$ since $\det(A) \neq 0, \det(B) \neq 0$.

(ii) Matrix multiplication is associative.

(iii) The $n \times n$ identity matrix I_n is the identity of the group.

(iv) Since $\det(A) \neq 0$ for all $A \in S$, we have that (from MA114) A has an inverse.

Therefore S forms a group under matrix multiplication.
Note that $AB \neq BA$ in general, so S is not an abelian group

Definition 3.5 (General linear group)

The group of all non-singular $n \times n$ matrices with real entries (under matrix multiplication) is called the *general linear group of degree n* and is denoted $GL(n, \mathbb{R})$.

Example 3.6

Consider the set of positive rational numbers $\mathbb{Q}^+$ together with the binary operation $*$ defined by $a * b = a \cdot b/2$ (where $\cdot$ denotes multiplication of rational numbers). Then:

(i) $\mathbb{Q}^+$ is closed under $*$ since for $a, b \in \mathbb{Q}^+$ we have $a * b \in \mathbb{Q}^+$.

(ii) $*$ is associative:
$(a * b) * c = (a \cdot b/2) * c = (a \cdot b/2) \cdot (c/2) = a \cdot b \cdot c/4$
$a * (b * c) = a * (b \cdot c/2) = (a \cdot (b \cdot c/2)/2) = a \cdot b \cdot c/4$
so $(a * b) * c = a * (b * c)$.

(iii) 2 is an identity element since $a * 2 = a$, $2 * a = a$ for all $a \in \mathbb{Q}^+$.

(iv) given $a \in \mathbb{Q}^+$, let $a' = 4/a$. Then $a * a' = a * (4/a) = (a \cdot (4/a))/2 = 2$, which is the identity; similarly $a' * a = 2$.

Observe here that the identity element isn't always what you might expect it to be.

Example 3.7 (Trivial group)

Let $G = \{e\}$ with multiplication defined by $e * e = e$. Then:

(i) $e * e = e \in G$ so G is closed;

(ii) $e * (e * e) = (e * e) * e$ so $*$ is associative;

(iii) the identity is e;

(iv) the inverse of e is e.

Therefore G is a group.

Definition 3.8 (Trivial group)

The group G consisting of a single element e, with multiplication defined by $e * e = e$ is called the *trivial group.*

Definition 3.9 (Group order)

Given a group G with operation $*$. The *order* of G is the number of elements in G (which may be infinite). We write $|G|$ to denote the order of G.

Example 3.10 (Group orders)

The groups in Example 3.3(a),(e),(f) and in Examples 3.4,3.6 are infinite groups (we write $|G| = \infty$).
The group in Example 3.3(h) has order 4 (written $|G| = 4$).
The group in Example 3.3(i) has order 2 ($|G| = 2$).
The trivial group has order 1 ($|G| = 1$).

Example 3.11 (Klein 4-group)

Let $G = \{e, a, b, c\}$ and let $*$ be defined by the table below. We shall show that G is a group.

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(i) multiplying any two elements of G gives another element of G, so G is closed under $*$.

(ii) It can be shown (for example by Light's associativity test – see Exercise 2.12) that $*$ is associative; for illustration we will only do two cases:

$$a * (b * c) = a * a = e$$
$$(a * b) * c = c * c = e$$

and

$$b * (c * b) = b * a = c$$
$$(b * c) * b = a * b = c.$$

(iii) The identity element is e.

(iv) Each element is its own inverse. This group is called the *Klein 4-group.*

Having given examples, we now begin to develop the theory of groups.
In some settings we cannot cancel terms in the way we might hope to, for example if X, Y, Z are matrices

$$XY = XZ \not\Rightarrow Y = Z$$

(for instance if $X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, Y = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, Z = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$). However, groups are a setting where we can cancel freely.

Lemma 3.12 (Left and right cancellation)

Let G be a group with binary operation $*$ and identity e and let $a, b, c \in G$.
If $a * b = a * c$ then $b = c$ and if $b * a = c * a$ then $b = c$.

Proof.

We prove the first part only, the second part being similar.

$$\begin{aligned} a * b = a * c &\Rightarrow a^{-1} * (a * b) = a^{-1} * (a * c) \quad \text{by the existence of } a^{-1} \\ &\Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c \quad \text{by associativity of } * \\ &\Rightarrow e * b = e * c \quad \text{by definition of inverse} \\ &\Rightarrow b = c \quad \text{by definition of identity element} \end{aligned}$$

□

Lemma 3.13

Let G be a group with binary operation $*$ and let $a, b \in G$. Then the linear equation $a * x = b$ has a unique solution for $x \in G$. Similarly, the linear equation $y * a = b$ has a unique solution for $y \in G$.

Proof.

We prove only the first case, leaving the second case as an exercise (Exercise 3.22.)
First note that $x = a^{-1} * b$ is a solution, since $a * x = a * (a^{-1} * b) = (a * a^{-1}) * b = e * b = b$.
We show that this is the only solution.
Suppose, for contradiction, that x_1, x_2 are each solutions with $x_1 \neq x_2$.
That is, $a * x_1 = b$ and $a * x_2 = b$; then $a * x_1 = a * x_2$ so by left cancellation (Lemma 3.12) we have $x_1 = x_2$, the required contradiction. □

Corollary 3.14

Let G be a finite group. Then in the multiplication table for G, each element $g \in G$ appears in each row exactly once, and appears in each column exactly once.

Proof.

The element $g \in G$ appears in row 'a' if and only if $a * b = g$ for some $b \in G$. This has a unique solution for b by Lemma 3.13.
The element $g \in G$ appears in column 'b' if and only if $a * b = g$ for some $a \in G$. This has a unique solution for a by Lemma 3.13. □

Lemma 3.15 (Unique identity)

Let G be a group with binary operation $*$. If e is an identity of G and f is an identity of G then $e = f$.

Proof.

Since e is an identity in G for all $x \in G$ we have that $e * x = x$. In particular, $f \in G$ so $e * f = f$.
Since f is an identity in G for all $x \in G$ we have that $x * f = x$. In particular, $e \in G$ so $e * f = e$.
Therefore $e = e * f = f$. □

Lemma 3.16 (Unique inverse)

Let G be a group with binary operation $*$ and identity e and let $a \in G$. If $a * b = b * a = e$ and $a * c = c * a = e$ then $b = c$. (That is, a has exactly one inverse element.)

Proof.

Exercise 3.23. □

Lemma 3.17 (Inverse of a product)

Let G be a group with binary operation $*$ and identity e and let $a, b \in G$. Then $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof.

We have

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = a * a^{-1} = e$$

and

$$(b^{-1} * a^{-1}) * (a * b) = b^{-1} * (a^{-1} * a) * b = b^{-1} * e * b = b^{-1} * b = e$$

so $b^{-1} * a^{-1}$ is an inverse of $a * b$.
By Lemma 3.16 this is the only inverse of $a * b$. □

Exercise 3.18

For each of the following determine whether the binary operation $*$ together with the given set form a group

(a) $*$ defined on $\mathbb{Z}$ by $a * b = ab$ (the product of a and b);

(b) $*$ defined on $2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\}$ by $a * b = a + b$ (the sum of a and b);

(c) $*$ defined on $\mathbb{Q}$ by $a * b = ab$ (the product of a and b);

(d) $*$ defined on $\mathbb{C}$ by $a * b = |ab|$ (i.e. the modulus of $|ab|$).

Exercise 3.19

Show that the set $S = \{1, (-1 + i\sqrt{3})/2, (-1 - i\sqrt{3})/2\}$ $(i = \sqrt{-1})$ together with the operation of multiplication of complex numbers forms a group.

Exercise 3.20

Recall that a *diagonal matrix* is square matrix whose only non-zero elements lie on the main diagonal (from the upper left corner to the lower right corner). Determine which of the following sets of diagonal matrices form groups under the given operation.

(a) All $n \times n$ diagonal matrices under matrix addition.

(b) All $n \times n$ diagonal matrices under matrix multiplication.

(c) All $n \times n$ diagonal matrices with all diagonal entries non-zero, under matrix multiplication.

(d) All $n \times n$ diagonal matrices whose diagonal entries are either 1 or -1, under matrix multiplication.

Exercise 3.21

Show that the set of 2×2 matrices with real entries and with determinant equal to 1 or -1 forms a group under matrix multiplication.

Exercise 3.22

Let G be a group with binary operation $*$. Show that the linear equation $y * a = b$ has a unique solution for $y \in G$.

Exercise 3.23

Let G be a group with identity e and let $g \in G$. Let $h, k \in G$ satisfy $gh = hg = e$ and $gk = kg = e$. Prove that $h = k$.

Exercise 3.24

Let G be a group in which $(ab)^2 = a^2b^2$ for all $a, b \in G$. Show that G is abelian.

Exercise 3.25

Let G be a group with identity e and suppose that $a, b, c \in G$ satisfy $abc = e$. Show that $bca = e$ in G.

Exercise 3.26

Let G be a group with identity e and let $a, b, c \in G$. Simplify $a(a^{-1}b)^{-1}(a^{-1}c^{-3}b)^{-1}a^{-1}c$.

Exercise 3.27

The table below is a partially completed multiplication table for a group.

$*$	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
g_0	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
g_1	g_1	g_0	g_3	g_2	g_5	g_4	g_7	g_6	g_9	g_8	g_{11}	g_{10}
g_2	g_2	g_3	g_0	g_1	g_6		g_4	g_5		g_{11}	g_8	g_9
g_3	g_3	g_2	g_1	g_0	g_7	g_6	g_5	g_4	g_{11}	g_{10}	g_9	g_8
g_4	g_4	g_5	g_8	g_9	g_0	g_1	g_{10}	g_{11}	g_2	g_3	g_6	g_7
g_5	g_5	g_4	g_9	g_8	g_1	g_0	g_{11}	g_{10}	g_3	g_2	g_7	g_6
g_6	g_6	g_7				g_3	g_8	g_9	g_0	g_1	g_4	g_5
g_7	g_7	g_6	g_{11}	g_{10}			g_9	g_8	g_1	g_0	g_5	g_4
g_8	g_8	g_9	g_4	g_5	g_{10}		g_0	g_1		g_7	g_2	g_3
g_9	g_9	g_8	g_5	g_4	g_{11}	g_{10}	g_1	g_0	g_7	g_6	g_3	g_2
g_{10}	g_{10}	g_{11}	g_6	g_7	g_8	g_9	g_2	g_3	g_4	g_5	g_0	g_1
g_{11}	g_{11}	g_{10}	g_7	g_6	g_9	g_8	g_3	g_2	g_5	g_4	g_1	g_0

Complete the missing entries.

Definition 4.1 (Subset)

A set B is a *subset* of a set A if every element of B is in A. We write $B \subseteq A$ (or $A \supseteq B$). If B is a subset of A and $B \neq A$ then B is a *proper subset* of A, and we write $B \subset A$ (or $A \supset B$).

Example 4.2 (Subset)

(a) $\mathbb{Z}$ is a proper subset of $\mathbb{R}$.

(b) $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ (the set of even integers) is a proper subset of $\mathbb{Z}$. The set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ is a proper subset of $\mathbb{Z}$.

(c) $\{-\pi, 15, 127\}$ is a subset of $\mathbb{R}$, but not of $\mathbb{Z}$.

(d) The set $\{1, 2, 3\}$ has 8 subsets, namely $\emptyset = \{\}$ (the empty set), $\{1\}, \{2\}, \{3\}, \{1, 2\}$, $\{1, 3\}, \{2, 3\}, \{1, 2, 3\}$. All except $\{1, 2, 3\}$ are proper subsets.

We now introduce the corresponding notion for groups. Now we need to also take account of the multiplication operation.

Definition 4.3 (Subgroup)

Let G be a group with operation $*$, and let H be a subset of G.
If H forms a group under the operation $*$ then we say that H is a *subgroup* of G, and write $H \leq G$.
If we have that $H \neq G$ then we write $H < G$ and say that H is a *proper subgroup* of G.

Example 4.4 (Subgroups of $\mathbb{Z}$)

Let $G = \mathbb{Z}$ be the group of integers under addition.

(a) The subset $H = 2\mathbb{Z}$ of even integers forms a subgroup of G since H forms a group under addition: (i) adding two even integers produces an even integer, (ii) addition of integers is associative, (iii) the number $0 \in H$ is the identity, (iv) the inverse of $2a \in H$ is $-2a \in H$.

(b) The subset $H = \{\ldots, -3, -1, 1, 3, \ldots\}$ of odd integers does not form a subgroup of G. For example, $1 + 3 \notin H$.

(c) The subset $H = \mathbb{N} = \{1, 2, 3, \ldots\}$ of natural numbers does not form a subgroup of G since, for example, it does not contain 0 (the identity of G).

(d) The subset $H = \{0, 1, 2, 3, \ldots\}$ of G does not form a subgroup of G since, while the identity $0 \in H$, the element 3 (for example), does not have an inverse in H (the inverse would be -3).

Example 4.5 (Subgroups of the Klein 4-group)

Recall (from Example 3.11) that the Klein 4-group G is the group with elements $\{e, a, b, c\}$ and multiplication $*$ defined by the table below.

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(i) Let $H = \{e, a\}$. Then

- H is closed under $*$;
- $*$ is associative;
- the identity $e \in H$;
- the inverse of e is $e \in H$ and the inverse of a is $a \in H$;

so H is a subgroup of G.

(ii) Let $H = \{a, b, c\}$. Then H is not a subgroup of G since (for example) $b * b = e \notin H$.

Example 4.6

Recall that the set of complex numbers $G = \{1, -1, i, -i\}$ under multiplication of complex numbers forms a group.
Let $H = \{1, -1\}$.
Then H forms a subgroup of G (it is easy to check that the four group properties (i),(ii),(iii),(iv) hold for H).
However $H = \{1, i\}$ is not a subgroup of G since, for example, $i \cdot i = -1 \notin H$.

Note that, when testing if a given subset H of a group G forms a subgroup of G, it is only necessary to test conditions (i),(iii),(iv), and not the associativity criterion, since if the binary operation is associative for G, then it is also associative for H.

Example 4.7 (The trivial group is a subgroup)

Let G be a group with identity e and let $H = \{e\}$. Then

(i) H is closed under multiplication;

(iii) the identity $e \in H$;

(iv) the inverse of $e \in H$ is $e \in H$;

so H is a subgroup of G. Thus the trivial group is a subgroup of any group G.

Exercise 4.8

Let G be the group of complex number $\mathbb{C}$ under addition of complex numbers. For each of the following, determine if the subset given forms a subgroup of G: (a) $\mathbb{R}$, (b) $\mathbb{N}$, (c) $3\mathbb{Z}$, (d) the set of purely imaginary numbers, $i\mathbb{R}$ (where $i^2 = -1$).

Exercise 4.9

Let $G = GL(n, \mathbb{R})$ (i.e. G is the group of $n \times n$ non-singular matrices with real entries under multiplication of matrices). For each of the following, determine if the subset given forms a subgroup of G:

(a) all $n \times n$ matrices with real entries and determinant 2;

(b) all $n \times n$ matrices with real entries and determinant -1;

(c) all $n \times n$ matrices with real entries and determinant 1 or -1;

(d) all $n \times n$ matrices with real entries and determinant 1 or -1 or 2 or -2.

Exercise 4.10

Let $= GL(n, \mathbb{R})$ (as defined in Exercise 4.9). A matrix $A \in GL(n, \mathbb{R})$ is *orthogonal* if $A^T A = I_n$, where A^T denotes the transpose of A and I_n is the $n \times n$ identity matrix. (Recall that for matrices A, B we have $(AB)^T = B^T A^T$ and $(A^{-1})^T = (A^T)^{-1}$.) Show that the subset of G consisting of all orthogonal matrices forms a subgroup of G.

Exercise 4.11

Consider the nine groups listed below. Find 13 subgroup relations of the form $G_i \leq G_j$ (where $G_i \neq G_j$) amongst them.

$G_1 = \mathbb{Z}$ under addition

$G_2 = 12\mathbb{Z}$ under addition

$G_3 = \mathbb{Q}^+$ under multiplication

$G_4 = \mathbb{R}$ under addition

$G_5 = \mathbb{R}^+$ under multiplication

$G_6 = \{\pi^n \mid n \in \mathbb{Z}\}$ under multiplication

$G_7 = 3\mathbb{Z}$ under addition

$G_8 =$ the set of all integral multiples of 6 under addition

$G_9 = \{6^n \mid n \in \mathbb{Z}\}$ under multiplication

Section 5. Cyclic groups.

Theorem 5.1

Let G be a group with identity e and let $a \in G$ and let

$$H = \{a^n \mid n \in \mathbb{Z}\} = \{\ldots, a^{-3}, a^{-2}, a^{-1}, a^0, a, a^2, a^3, \ldots\}$$

Then H is a subgroup of G and every subgroup of G that contains a contains H.

Proof.

As remarked after Example 4.6, to show that H is a subgroup we must test if properties (i),(iii),(iv) of Definition 3.1 hold.

(i) If $a^n, a^m \in H$ then $a^n a^m = a^{n+m} \in H$.

(iii) Note that $a^1 a^{-1} = e$ and $a^1 a^{-1} = a^{1-1} = a^0$ so $a^0 \in G$ is the identity of G; further $a^0 \in H$ so $e \in H$.

(iv) If $a^r \in H$ then $a^{-r} \in H$ and $a^r a^{-r} = a^0 = e$, $a^{-r} a^r = a^0 = e$ so a^{-r} is the inverse of a^r.

Let K be a subgroup of G that contains a. Then since K is closed under $*$, it must contain all powers of a. That is, K must contain H. □

Definition 5.2 (Cyclic group)

Let G be a group and let $a \in G$.

(a) The group $H = \{a^n \mid n \in \mathbb{Z}\}$ is called the *cyclic subgroup of G generated by a*, and is denoted by $< a >$.

(b) If $< a >= G$ then we say that a *generates* G or that a is a *generator of* G.

(c) A group G is *cyclic* if there is some $a \in G$ that generates G.

(d) The *order* of a is the order $|H|$ (i.e. the number of elements of H).

Example 5.3 (Cyclic subgroups of the Klein 4 group)

Consider the Klein 4-group G, i.e. the group with elements $\{e, a, b, c\}$ and multiplication $*$ defined by the table below.

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(Note, in particular, that e is the identity.)
The set

$$\{a^n \mid n \in \mathbb{Z}\} = \{\ldots, a^{-3}, a^{-2}, a^{-1}, a^0, a^1, a^2, \ldots\}.$$

But $a^2 = e$ and $a = a^{-1}$ so all even powers of a are equal to e and all odd powers of a are equal to a.
Therefore $< a >= \{a^n \mid n \in \mathbb{Z}\} = \{e, a\}$, i.e. the cyclic subgroup of G generated by a is $\{e, a\}$.
Therefore a has order 2, and a does not generate G.
Similarly b, c each have order 2, and do not generate G.
The set $< e >= \{e^n \mid n \in \mathbb{Z}\} = \{e\}$ so e has order 1, and does not generate G.
Therefore G is not cyclic.

Example 5.4 (Cyclic subgroups of a group of four complex numbers)

Recall that the set $G = \{1, -1, i, -i\}$ under multiplication of complex numbers forms a group.
The cyclic subgroup of G generated by -1 is $< -1 >= \{(-1)^n \mid n \in \mathbb{Z}\} = \{1, -1\}$, so -1 has order 2, and does not generate G.
The cyclic subgroup of G generated by i is $< i >= \{i^n \mid n \in \mathbb{Z}\} = \{1, i, -1, -i\} = G$, so i has order 4 and generates G, so G is cyclic.

Example 5.5 (Cyclic subgroups of $\mathbb{Z}$)

Recall that $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ forms a group under addition.
The cyclic subgroup of G generated by 2 is

$$< 2 >= \{\ldots, -4, -2, 0, 2, 4, \ldots\} = 2\mathbb{Z} \neq \mathbb{Z}$$

so 2 has infinite order but does not generate $\mathbb{Z}$.
The cyclic subgroup of G generated by -1 is

$$< -1 >= \{\ldots, 2, 1, 0, -1, -2, \ldots\} = \mathbb{Z},$$

so -1 has infinite order and generates $\mathbb{Z}$ so $\mathbb{Z}$ is cyclic.
Similarly the cyclic subgroup of G generated by 1 is $\{\ldots, -2, -1, 0, 1, 2, \ldots\} = \mathbb{Z}$, so 1 generates $\mathbb{Z}$. In particular, $\mathbb{Z}$ is cyclic.

Definition 5.6

Let $n \geq 2$. The group $\mathbb{Z}_n$ consists of the set of elements $\{0, 1, 2, \ldots, n-1\}$ together with a binary operation '$+$' where $a + b$ is defined to be the remainder when the sum of a and b (taken as integers) is divided by n. (Thus this remainder is an element of the set $\{0, 1, \ldots, n-1\}$.) This operation is called *addition modulo n*.

Lemma 5.7

$\mathbb{Z}_n$ (under addition modulo n) is a cyclic group.

Proof.

(i) Remainders of an integer divided by n belong to $\{0, 1, \ldots, n-1\}$ so $\mathbb{Z}_n$ is closed under $+$.

(ii) Addition is associative.

(iii) The identity is 0.

(iv) The inverse of $i \in \{1, \ldots, n-1\}$ is $n - i \in \{1, \ldots, n-1\}$; the inverse of 0 is 0.

Thus $\mathbb{Z}_n$ is a group.
Every element of $\mathbb{Z}_n$ can be written as a power of the element 1, so 1 is a generator of $\mathbb{Z}_n$, so $\mathbb{Z}_n$ is cyclic. □

Example 5.8

$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ and has multiplication table below.

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Example 5.9 (Trivial group is cyclic)

Let G be the trivial group, $G = \{e\}$. Then $e \in G$ is a generator for G, so G is cyclic.

Exercise 5.10

Write down the tables for the groups $\mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_4, \mathbb{Z}_6$.

Exercise 5.11

Is the group $\mathbb{Z}_n$ abelian? Justify your answer.

Exercise 5.12

Find all generators of the cyclic group $\mathbb{Z}_{12}$.

Exercise 5.13

Give an example of a non-cyclic group G, all of whose proper subgroups are cyclic.

Exercise 5.14

Let G be a group of order 3. Show that G is cyclic.

Exercise 5.15

Let H be the cyclic subgroup of $\mathbb{C}^*$ generated by $(-1 + i\sqrt{3})/2$. Find the order of H.

Exercise 5.16

Let G be a group that has precisely one element of order 2, h (say). Show that h commutes with every element $g \in G$. (Hint: consider the element ghg^{-1}.)

Exercise 5.17

Let g be an element of a group G. Show that the order of g is equal to the order of g^{-1}. Deduce that no group has a unique element of order 3.

Exercise 5.18

Let G be a group in which every non-trivial element has order 2. Show that G is abelian.

Section 6. Groups of permutations.

Definition 6.1 (Function, mapping, one-one, onto)

A *function* or *mapping* $\phi : A \to B$ from a set A to a set B is a rule that assigns to each element a of A one element b of B.
A function $\phi : A \to B$ is *one-to-one* if $\phi(a_1) = \phi(a_2)$ implies that $a_1 = a_2$.
A function $\phi : A \to B$ is *onto* if for all $b \in B$ there exists $a \in A$ such that $\phi(a) = b$.

Example 6.2

(a) Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 2x + 3$.
Then for all $y \in \mathbb{R}$ we have that $f((y-3)/2) = y$ so f is onto.
If $f(x) = f(y)$ then $2x + 3 = 2y + 3$ and so $x = y$, so f is one-one.

(b) Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$.
Then $f(x) \geq 0$ so f is not onto (for example there is no $x \in \mathbb{R}$ such that $f(x) = -1$).
Also $f(x) = f(-x)$ so f is not one-one.

(c) Consider the function $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(x) = 2x$.
Then f is not onto since, for example, there is no $x \in \mathbb{Z}$ such that $f(x) = 1$.
However $f(x) = f(y)$ implies $2x = 2y$ so $x = y$, and hence f is one-one.

We now turn to our main object of study in the section, namely permutations.

Definition 6.3

A *permutation of a set* A is a function $\phi : A \to A$ that is both one-one and onto. We can write permutations in *two-line* format, a two-line tabular format where for each $a \in A$ the element a is written directly above $\phi(a)$.

Example 6.4

Let $A = \{1, 2, 3, 4, 5\}$, and let $\phi : A \to A$ be given by

$$\phi(1) = 4, \phi(2) = 2, \phi(3) = 5, \phi(4) = 3, \phi(5) = 1.$$

Then clearly ϕ is one-one and onto, so ϕ is a permutation.
In two-line format this is

$$\phi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}.$$

Lemma 6.5 (Composition of permutations is a permutation)

Let σ, τ be permutations of a set A. Then $\sigma \circ \tau$ is a permutation of A.

Proof.

We have that $\sigma : A \to A$ is one-one and onto, and $\tau : A \to A$ is one-one and onto.
Let $\mu = \sigma \circ \tau$. Then $\mu : A \to A$ is given by $\mu(a) = \sigma(\tau(a))$. We claim that μ is one-one and onto, and so μ is a permutation.
Suppose for contradiction that μ is not one-one, i.e. for some $a \neq b \in A$ we have $\mu(a) = \mu(b)$. Then $\sigma(\tau(a)) = \sigma(\tau(b))$; let $a' = \tau(a), b' = \tau(b)$, and so $\sigma(a') = \sigma(b')$. Now τ is one-one so $a' \neq b'$, and hence (since σ is one-one) we have $\sigma(a') \neq \sigma(b')$, a contradiction. Therefore μ is one-one.
We now show that μ is onto. Let $a \in A$.
Now σ is onto, so there exists $b \in A$ such that $\sigma(b) = a$.
Since τ is onto, there exists $c \in A$ such that $\tau(c) = b$.
Thus $\mu(c) = (\sigma \circ \tau)(c) = \sigma(\tau(c)) = \sigma(b) = a$, hence μ is onto. □

In order to have a group in which the group elements are permutations we need to be able to multiply permutations.

Definition 6.6

Let σ, τ be permutations of a set A. We define their product $\sigma * \tau$ to be the permutation $\sigma \circ \tau$.

(Note that by Lemma 6.5 $\sigma * \tau$ is a permutation so $*$ is a closed binary operation on the set of all permutations of a set A.)

Example 6.7

Let $A = \{1, 2, 3, 4, 5\}$ and let σ, τ be the following permutations

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}, \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}.$$

Then

$$\sigma(1) = 4, \sigma(2) = 2, \sigma(3) = 5, \sigma(4) = 3, \sigma(5) = 1,$$
$$\tau(1) = 3, \tau(2) = 5, \tau(3) = 4, \tau(4) = 2, \tau(5) = 1$$

so

$$\begin{aligned} \sigma(\tau(1)) &= \sigma(3) = 5, \\ \sigma(\tau(2)) &= \sigma(5) = 1, \\ \sigma(\tau(3)) &= \sigma(4) = 3, \\ \sigma(\tau(4)) &= \sigma(2) = 2, \\ \sigma(\tau(5)) &= \sigma(1) = 4. \end{aligned}$$

Thus (in two-line notation) the product $\sigma * \tau$ is

$$\sigma * \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix}.$$

Example (Example 6.7 continued)

There is no need to be this laborious in multiplying permutations once you know what you are doing (the above explanation was to reinforce the point that permutations are functions).
We could just write

$$\sigma * \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} * \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix}$$

(thinking τ takes 1 to 3, then σ takes 3 to 5, giving the first column of the answer; or that τ takes 2 to 5, then σ takes 5 to 1, giving the second column of the answer; etc).

Note that to calculate $\sigma * \tau$ we have to start by thinking about what τ does, then think about what σ does: this is sometimes called 'multiplying right to left'.

Let's do another example (this time we omit the '$*$' to denote the product – we will do this increasingly often if it is clear what the binary operation is):

Example 6.8

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}, \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 3 & 6 & 5 \end{pmatrix}.$$

Then

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 3 & 4 & 2 & 6 \end{pmatrix}$$

and

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix}.$$

(One check you can make is that each number should appear exactly once in the bottom line of your answer – if it doesn't then you have made a mistake, as it would mean that your answer is not a permutation.)

Theorem 6.9 (Symmetric group)

Let $A = \{1, 2, \ldots, n\}$ and let G be the set of all permutations of A. Then the set G together with permutation multiplication forms a group.

Proof.

(i) As shown in Lemma 6.5 permutation multiplication is a closed binary operation on the set of all permutations of a set A.

(ii) As shown in Example 2.6(c), composition of functions is associative; therefore multiplication of permutations is an associative binary operation.

(iii) The permutation $\iota : A \to A$ given by $\iota(a) = a$ for all $a \in A$ is the identity of G since $\iota \circ \sigma = \sigma$ and $\sigma \circ \iota = \sigma$ for all permutations σ of A.

(iv) By Exercise 6.17, for all permutations σ of A there exists a permutation τ such that $\sigma \circ \tau = \iota$ and $\tau \circ \sigma = \iota$, so τ is the inverse of σ.

□

Definition 6.10 (Symmetric group)

Let $A = \{1, 2, \ldots, n\}$ and let G be the set of all permutations of A. The group consisting of G together with permutation multiplication is called the *symmetric group* and is denoted S_n.

Example 6.11 (Order of the symmetric group)

Let $\sigma \in S_n$. Then there are n possible values of $\sigma(1)$; once this is fixed there are $n-1$ possible values of $\sigma(2)$; once this is fixed there are $n-2$ possible values of $\sigma(3)$; and so on.
Therefore there are $n!$ permutations of $\{1, 2, \ldots, n\}$, i.e. $|S_n| = n!$.

Example 6.12 (The symmetric group S_3 and symmetries of an equilateral triangle)

The symmetric group S_3 is the group of permutations of $A = \{1, 2, 3\}$, and so has $3! = 6$ elements, which we label as follows:

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \mu_a = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$
$$\rho_{120} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \qquad \mu_b = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix},$$
$$\rho_{240} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \qquad \mu_c = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Now we know how to multiply permutations, it is easy to find that the multiplication table is as given below:

	ρ_0	ρ_{120}	ρ_{240}	μ_a	μ_b	μ_c
ρ_0	ρ_0	ρ_{120}	ρ_{240}	μ_a	μ_b	μ_c
ρ_{120}	ρ_{120}	ρ_{240}	ρ_0	μ_c	μ_a	μ_b
ρ_{240}	ρ_{240}	ρ_0	ρ_{120}	μ_b	μ_c	μ_a
μ_a	μ_a	μ_b	μ_c	ρ_0	ρ_{120}	ρ_{240}
μ_b	μ_b	μ_c	μ_a	ρ_{240}	ρ_0	ρ_{120}
μ_c	μ_c	μ_a	μ_b	ρ_{120}	ρ_{240}	ρ_0

Note that S_3 is not abelian (e.g. $\rho_{120}\mu_a = \mu_c$ but $\mu_a\rho_{120} = \mu_b$).
We now show that the group S_3 describes the symmetries of an equilateral triangle.

Example (Example 6.12 continued)

Consider an equilateral triangle, with its base horizontal, with vertices labelled $1, 2, 3$ clockwise from the bottom left.

- If we apply an anticlockwise rotation by 120 degrees to this then:
  - vertex label 1 is replaced by 2;
  - vertex label 2 is replaced by 3;
  - vertex label 3 is replaced by 1.

  Therefore this rotation corresponds to the permutation

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \rho_{120}.$$

- If we apply a reflection in the line bisecting the angle at the top vertex to the original triangle then
  - vertex label 1 is replaced by 3;
  - vertex label 2 is replaced by 2;
  - vertex label 3 is replaced by 1.

  Therefore this rotation corresponds to the permutation

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \mu_b.$$

Example (Example 6.12 continued)

Similar considerations show that:

- ρ_0 corresponds to an anticlockwise rotation by 0 degrees;
- ρ_{120} corresponds to an anticlockwise rotation by 120 degrees;
- ρ_{240} corresponds to an anticlockwise rotation by 240 degrees;
- μ_a corresponds to reflection in the line bisecting the angle at the bottom left vertex;
- μ_b corresponds to reflection in the line bisecting the angle at the top vertex;
- μ_c corresponds to reflection in the line bisecting the angle at the bottom right vertex.

Then, for example $\mu_c\rho_{120}$ corresponds to reflecting in the line bisecting the angle at the bottom right vertex followed by an anticlockwise rotation by 120 degrees. Doing this leaves label 1 in the bottom left corner, label 2 in the bottom right and label 3 at the top.
This reflection followed by rotation corresponds to the permutation

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \mu_a.$$

So we should have $\mu_c\rho_{120} = \mu_a$. Checking against the multiplication table, we see that this is indeed the case.

Example (Example 6.12 continued)

Similarly $\rho_{240}\rho_{120}$ corresponds to an anticlockwise rotation by 240 degrees followed by an anticlockwise rotation by 120 degrees. This therefore corresponds to

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \rho_0.$$

So we should have $\rho_{240}\rho_{120} = \rho_0$. Checking against the multiplication table we see that this is indeed the case.
Try out some examples of your own.

The group S_3 is the group of symmetries of the regular 3-gon, and has order $6 = 2 \cdot 3$.
The group of symmetries of the regular n-gon ($n \geq 3$) has $2n$ elements is called the *dihedral group of order* $2n$, denoted D_{2n}.
Therefore S_3 is the dihedral group D_6. In the next example we study D_8.

Example 6.13 (The group D_8 and symmetries of the square)

A regular 4-gon is a square. Consider a square on the page, with horizontal base, and vertices labelled $1, 2, 3, 4$ clockwise from the bottom left. Let

- Let $\rho_0, \rho_{90}, \rho_{180}, \rho_{270}$ be anticlockwise rotations of the square by angles $0, 90, 180, 270$ degrees, respectively;
- let μ_H be the reflection in a horizontal line passing through the midpoints of the vertical edges of the square;
- let μ_V be the reflection in a vertical line passing through the midpoints of the horizontal edges of the square;
- let μ_U be reflection in diagonal passing through the South West and North East corners, (the 'upward diagonal');
- and let μ_D be reflection in diagonal passing through the North West and South East corners, (the 'downward diagonal').

Then the action of these rotations and reflections corresponds to the following permutations:

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \quad \mu_H = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix},$$
$$\rho_{90} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \quad \mu_V = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix},$$
$$\rho_{180} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \quad \mu_D = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix},$$
$$\rho_{270} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}, \quad \mu_U = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}.$$

Example (Example 6.13 continued)

These permutations form the group D_8, which therefore has the following multiplication table:

	ρ_0	ρ_{90}	ρ_{180}	ρ_{270}	μ_H	μ_V	μ_D	μ_U
ρ_0	ρ_0	ρ_{90}	ρ_{180}	ρ_{270}	μ_H	μ_V	μ_D	μ_U
ρ_{90}	ρ_{90}	ρ_{180}	ρ_{270}	ρ_0	μ_D	μ_U	μ_V	μ_H
ρ_{180}	ρ_{180}	ρ_{270}	ρ_0	ρ_{90}	μ_V	μ_H	μ_U	μ_D
ρ_{270}	ρ_{270}	ρ_0	ρ_{90}	ρ_{180}	μ_U	μ_D	μ_H	μ_V
μ_H	μ_H	μ_U	μ_V	μ_D	ρ_0	ρ_{180}	ρ_{270}	ρ_{90}
μ_V	μ_V	μ_D	μ_H	μ_U	ρ_{180}	ρ_0	ρ_{90}	ρ_{270}
μ_D	μ_D	μ_H	μ_U	μ_V	ρ_{90}	ρ_{270}	ρ_0	ρ_{180}
μ_U	μ_U	μ_V	μ_D	μ_H	ρ_{270}	ρ_{90}	ρ_{180}	ρ_0

Then, for example, reflection μ_V followed by rotation ρ_{270} corresponds to the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} = \mu_U$$

So we should have $\mu_V \rho_{270} = \mu_U$. Checking against the table we see that this is indeed the case. Try some examples of your own.

Exercise 6.14

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix},$$
$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 3 & 6 & 5 \end{pmatrix},$$
$$\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 4 & 3 & 1 & 6 \end{pmatrix}.$$

Compute the following products: (i) $\tau\sigma$ (ii) $\tau^2\sigma$ (iii) $\mu\sigma^2$ (iv) $\sigma^{-2}\tau$ (v) $\sigma^{-1}\tau\sigma$

Exercise 6.15

(a) Find the number of elements in the set $H = \{\sigma \in S_3 \mid \sigma(3) = 1\}$. Is H a subgroup of S_3? (Justify your answer.)

(b) Find the number of elements in the set $H = \{\sigma \in S_5 \mid \sigma(1) = 1\}$. Is H a subgroup of S_5? (Justify your answer.)

Exercise 6.16

Consider the symmetric group $S_3 = \{\rho_0, \rho_{120}, \rho_{240}, \mu_a, \mu_b, \mu_c\}$ of Example 6.12. Let $H =< \rho_{120} >$, $K =< \mu_a >$ be cyclic subgroups of S_3. Find all the elements of H and find all the elements of K.

Exercise 6.17

Let $\sigma : A \to B$ be one-one and onto. Show that there exists $\tau : B \to A$ such that $\tau \circ \sigma(a) = a$ for all $a \in A$ and $\sigma \circ \tau(b) = b$ for all $b \in B$.

Section 7. Cosets and Lagrange's Theorem

Cosets are a way of breaking a group into smaller pieces to help us understand the structure of a group. They allow us to prove an important result, Lagrange's Theorem, which relates the order of a group to the order of a subgroup, which has the consequence (for example) that any group of prime order is cyclic. In MA301 Group Theory (if you are taking it) you will learn that cosets can be used to form new groups (called *quotient groups*) and these provide a deep structural understanding of the group. In MA315 Cryptography and Codes (if you are taking it) you will learn that applications of cosets include error correction codes.

Theorem 7.1 (Lagrange's Theorem)

Let G be a finite group, H a subgroup of G. Then the order $|H|$ divides the order $|G|$.

We will prove Lagrange's Theorem after we have developed the theory of cosets that will allow us to do so.

Corollary 7.2 (Groups of prime order)

Every group of prime order is cyclic.

Proof.

Let G be a group with $|G| = p$, where $p \geq 2$ is prime, and let $a \in G$ be an element of G different to the identity e. Let $H =< a >$ be the cyclic subgroup of G generated by a. Then H has at least two elements, namely a and the identity e. By Theorem 7.1 we have $|H|$ divides p, so $|H| = 1$ or p. But $|H| \neq 1$, so $|H| = p$. Hence all of the elements of G are elements of H so $G \subseteq H$, and $H \leq G$ so $H = G$. Then since H is cyclic, the group G is cyclic. □

Recall from Definition 5.2(d) that the order of an element g of a group G, denoted $|g|$, is the number of elements in the cyclic subgroup of G generated by g.

Corollary 7.3 (Order of a group element)

Let G be a finite group and let $g \in G$. Then $|g|$ divides $|G|$.

Proof.

Let $H =< g >$. Then $|g| = |H|$, and by Theorem 7.1 we have $|H|$ divides $|G|$ so $|g|$ divides $|G|$. □

Example 7.4

(a) In a group of order 12, the possible orders of group elements are $1, 2, 3, 4, 6, 12$.

(b) In a group of order 4, the possible orders of group elements are $1, 2, 4$. Note that not all of these values have to occur, for example the Klein 4 group has no element of order 4.

Definition 7.5

Let G be a group and let H be a subgroup of G.
For each $g \in G$ the set

$$gH = \{gh \mid h \in H\}$$

is called the *left coset of H containing g* and the set

$$Hg = \{hg \mid h \in H\}$$

is called the *right coset of H containing g*.

Example 7.6

Let $G = S_3 = \{\rho_0, \rho_{120}, \rho_{240}, \mu_a, \mu_b, \mu_c\}$ where these permutations and the multiplication table are given below:

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mu_a = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$
$$\rho_{120} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad \mu_b = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix},$$
$$\rho_{240} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \quad \mu_c = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

	ρ_0	ρ_{120}	ρ_{240}	μ_a	μ_b	μ_c
ρ_0	ρ_0	ρ_{120}	ρ_{240}	μ_a	μ_b	μ_c
ρ_{120}	ρ_{120}	ρ_{240}	ρ_0	μ_c	μ_a	μ_b
ρ_{240}	ρ_{240}	ρ_0	ρ_{120}	μ_b	μ_c	μ_a
μ_a	μ_a	μ_b	μ_c	ρ_0	ρ_{120}	ρ_{240}
μ_b	μ_b	μ_c	μ_a	ρ_{240}	ρ_0	ρ_{120}
μ_c	μ_c	μ_a	μ_b	ρ_{120}	ρ_{240}	ρ_0

Let $H = \{\rho_0, \mu_a\}$. Then the left cosets of H are

$$\rho_0 H = \{\rho_0\rho_0, \rho_0\mu_a\} = \{\rho_0, \mu_a\} = H, \quad \mu_a H = \{\mu_a\rho_0, \mu_a\mu_a\} = \{\mu_a, \rho_0\},$$
$$\rho_{120} H = \{\rho_{120}\rho_0, \rho_{120}\mu_a\} = \{\rho_{120}, \mu_c\}, \quad \mu_b H = \{\mu_b\rho_0, \mu_b\mu_a\} = \{\mu_b, \rho_{240}\},$$
$$\rho_{240} H = \{\rho_{240}\rho_0, \rho_{240}\mu_a\} = \{\rho_{240}, \mu_b\}, \quad \mu_c H = \{\mu_c\rho_0, \mu_c\mu_a\} = \{\mu_c, \rho_{120}\}.$$

Notice that $\mu_a H = \rho_0 H = H$, $\mu_b H = \rho_{240} H$, $\mu_c H = \rho_{120} H$, so there are three distinct left cosets.

Example (Example 7.6 continued)

The right cosets of H are

$$\begin{aligned} H\rho_0 &= \{\rho_0\rho_0, \mu_a\rho_0\} = \{\rho_0, \mu_a\} = H, & H\mu_a &= \{\rho_0\mu_a, \mu_a\mu_a\} = \{\mu_a, \rho_0\}, \\ H\rho_{120} &= \{\rho_0\rho_{120}, \mu_a\rho_{120}\} = \{\rho_{120}, \mu_b\}, & H\mu_b &= \{\rho_0\mu_b, \mu_a\mu_b\} = \{\mu_b, \rho_{120}\}, \\ H\rho_{240} &= \{\rho_0\rho_{240}, \mu_a\rho_{240}\} = \{\rho_{240}, \mu_c\}, & H\mu_c &= \{\rho_0\mu_c, \mu_a\mu_c\} = \{\mu_c, \rho_{240}\}. \end{aligned}$$

Notice that $H\mu_a = H\rho_0 = H$, $H\mu_b = H\rho_{120}$, $H\mu_c = H\rho_{240}$, so there are three distinct right cosets. The left cosets are not the same as the right cosets.

Example 7.7

Consider the group $\mathbb{Z}_6$ with elements $\{0, 1, 2, 3, 4, 5\}$ with the binary operation '$+$' defined in the table.

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Then the set $H = \{0, 3\}$ forms a subgroup of G (check you understand why). The left cosets of H are

$$\begin{aligned} 0 + H &= \{0, 3\} = H, & 3 + H &= \{3, 0\}, \\ 1 + H &= \{1, 4\}, & 4 + H &= \{4, 1\}, \\ 2 + H &= \{2, 5\}, & 5 + H &= \{5, 2\}. \end{aligned}$$

(We use the '$+$' notation for cosets here, since '$+$' is the symbol we gave to the binary operation.) Notice that $3 + H = H$, $4 + H = 1 + H$, $5 + H = 2 + H$, so there are 3 distinct left cosets.

Example (Example 7.7 continued)

The right cosets of H are

$$\begin{aligned} H+0 &= \{0,3\} = H, & H+3 &= \{3,0\}, \\ H+1 &= \{1,4\}, & H+4 &= \{4,1\}, \\ H+2 &= \{2,5\}, & H+5 &= \{5,2\}. \end{aligned}$$

In this case the set of left cosets is equal to the set of right cosets.

We now derive some important properties of cosets.

Lemma 7.8 (Properties of cosets)

Let H be a subgroup of a group G and let $a, b \in G$. Then

(a) *either $aH = bH$ or $aH \cap bH = \emptyset$,*

(b) $|aH| = |bH|$,

(c) $|Ha| = |Hb|$.

Proof.

(a) Suppose $aH \cap bH \neq \emptyset$.
We shall show $aH \subseteq bH$ and $bH \subseteq aH$ so $aH = bH$.
Since $aH \cap bH \neq \emptyset$ there exists $x \in aH \cap bH$. Then $x = ah_1$ and $x = bh_2$ for some $h_1, h_2 \in H$.
Then $a = xh_1^{-1} = bh_2h_1^{-1}$ and so for any $h \in H$ we have that $ah = b(h_2h_1^{-1}h) \in bH$.
Thus $aH \subseteq bH$
Further $b = xh_2^{-1} = ah_1h_2^{-1}$ and so for any $h \in H$ we have that $bh = a(h_1h_2^{-1}h) \in aH$.
Thus $bH \subseteq aH$. Hence $aH = bH$.

(b) We show that $|gH| = |H|$ for all $g \in G$. Therefore $|aH| = |H|$,$|bH| = |H|$, so $|aH| = |bH|$.
Let $g \in G$ and define $\phi : H \to gH$ by $\phi(h) = gh$. We show that ϕ is one-one and onto, so $|H| = |gH|$. Now $\phi(h_1) = \phi(h_2)$ implies that $gh_1 = gh_2$ so $h_1 = h_2$, so ϕ is one-one. Let $gh \in gH$. Then $gh = \phi(h)$, so ϕ is onto.

(c) Exercise 7.14.

□

Recall that Lagrange's Theorem (Theorem 7.1) states that if G is a finite group, and H is a subgroup of G then the order $|H|$ divides the order $|G|$. We are now in a position to prove this.

Proof of Lagrange's Theorem, Theorem 7.1.

Since G is a finite group it has finitely many distinct left cosets $g_1H, g_2H, \ldots, g_rH$, say.
Moreover, each $g \in G$ can be written $g = g \cdot e$, so is an element of the coset gH, so every element of G is in some coset, and hence $G \subseteq g_1H \cup g_2H \cup \ldots \cup g_rH \subseteq G$, so $G = g_1H \cup g_2H \cup \ldots \cup g_rH$.
By Lemma 7.8(a) we have $g_iH \cap g_jH = \emptyset$ for $i \neq j$, so $|G| = |g_1H| + |g_2H| + \ldots + |g_rH|$.
By Lemma 7.8(b) we have $|g_iH| = |eH|$, where $e \in G$ is the identity of G.
But $eH = H$ so $|g_iH| = |H|$. Thus $|G| = |H| + |H| + \ldots + |H|$ (r summands), so $|G| = r|H|$, and in particular, $|H|$ divides $|G|$. □

Corollary 7.9

(a) *The number of distinct left cosets of H in G is equal to $|G|/|H|$.*

(b) *The number of distinct right cosets of H in G is equal to $|G|/|H|$.*

Proof.

(a) The number of left cosets of H in G is the number r in the proof of Theorem 7.1. From that proof $|G| = r|H|$ so $r = |G|/|H|$.

(b) Suppose there are s distinct right cosets of H in G. Each pair of cosets has empty intersection by Lemma 7.8(a) and each has $|H \cdot e| = |H|$ elements by Lemma 7.8(c), where e is the identity of G. Thus $|G| = s \cdot |H|$ so $s = |G|/|H|$.

□

Definition 7.10

Let G be a finite group and let H be a subgroup of G. The *index* of H in G is the number of left cosets of H in G, and is denoted $[G : H]$.

Example 7.11

Let G be the symmetric group S_4 and let H be the subgroup of G consisting of the following permutations:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}.$$

Then the index $[G : H] = |G|/|H| = 4!/4 = 3! = 6$, so there are 6 left cosets of H in G and 6 right cosets of H in G.

Exercise 7.12

Let G be the group $\mathbb{Z}_{12}$. Let $H = \{0, 3, 6, 9\}$ be a subgroup of G. Find the left cosets of H in G.

Exercise 7.13

Let G be the group D_8 as given in Example 6.13.

(a) Consider the subgroup $H = \{\rho_0, \rho_{90}, \rho_{180}, \rho_{270}\}$ of G. Find the index $[G : H]$, find the left cosets of H in G, and find the right cosets of H in G.

(b) Consider the subgroup $H = \{\rho_0, \mu_U\}$ of G. Find the index $[G : H]$, find the left cosets of H in G, and find the right cosets of H in G.

Exercise 7.14

Let H be a subgroup of a group G and let $a, b \in G$. Show that $|Ha| = |Hb|$.

Exercise 7.15

Let G be a finite group of order pq where p, q are prime. Show that every proper subgroup of G is cyclic.

Section 8. Direct Products

Direct products provide a way of gluing groups together to get new groups.

Definition 8.1 (Cartesian product of sets)

Let A, B be sets. The *Cartesian product of A and B* is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

Example 8.2

(a) Let $A = \{x, y, z\}, B = \{p, q\}$ then
$A \times B = \{(x, p), (x, q), (y, p), (y, q), (z, p), (z, q)\}$.

(b) $\mathbb{Z} \times \mathbb{Z} = \{(x, y) \mid x \in \mathbb{Z}, y \in \mathbb{Z}\}$.

We now introduce a similar concept for groups – we will need to build the multiplication into the picture.

Theorem 8.3 (Direct product of groups)

Let G, H be groups with binary operations $*_G, *_H$. Define a binary operation $*$ on the cartesian product $G \times H$ by

$$(g_1, h_1) * (g_2, h_2) = (g_1 *_G g_2, h_1 *_H h_2).$$

Then $G \times H$ together with this binary operation is a group.

Proof.

(i) Let $(g_1, h_1), (g_2, h_2) \in G \times H$. Then

$$(g_1, h_1) * (g_2, h_2) = (g_1 *_G g_2, h_1 *_H h_2) \in G \times H$$

so $G \times H$ is closed under $*$.

(ii) Let $g_1, g_2, g_3 \in G$ and let $h_1, h_2, h_3 \in H$. Since G, H are groups we have that $*_G, *_H$ are associative, i.e. $(g_1 *_G g_2) *_G g_3 = g_1 *_G (g_2 *_G g_3)$ and $(h_1 *_H h_2) *_H h_3 = h_1 *_H (h_2 *_H h_3)$. Thus

$$\begin{aligned}((g_1, h_1) * (g_2, h_2)) * (g_3, h_3) &= (g_1 *_G g_2, h_1 *_H h_2) * (g_3, h_3) \\ &= ((g_1 *_G g_2) *_G g_3, (h_1 *_H h_2) *_H h_3) \\ &= (g_1 *_G (g_2 *_G g_3), h_1 *_H (h_2 *_H h_3)) \\ &= (g_1, h_1) * (g_2 *_G g_3, h_2 *_H h_3) \\ &= (g_1, h_1) * ((g_2, h_2) * (g_3, h_3))\end{aligned}$$

Proof of Theorem 8.3 continued.

(iii) Let e_G, e_H denote the identities of G, H, respectively.Then

$$(e_G, e_H) * (g, h) = (e_G *_G g, e_H *_H h) = (g, h)$$

and

$$(g, h) * (e_G, e_H) = (g *_G e_G, h *_H e_H) = (g, h)$$

so (e_G, e_H) is the identity of $G \times H$.

(iv) Let $(g, h) \in G \times H$. Then $g \in G, h \in H$ so have inverses $g^{-1} \in G, h^{-1} \in H$ and

$$(g, h) * (g^{-1}, h^{-1}) = (g *_G g^{-1}, h *_H h^{-1}) = (e_G, e_H)$$

and

$$(g^{-1}, h^{-1}) * (g, h) = (g^{-1} *_G g, h^{-1} *_H h) = (e_G, e_H)$$

so (g^{-1}, h^{-1}) is the inverse of (g, h).

□

Definition 8.4 (Direct product of groups)

Let G, H be groups with binary operations $*_G, *_H$. Define a binary operation $*$ on the cartesian product $G \times H$ by

$$(g_1, h_1) * (g_2, h_2) = (g_1 *_G g_2, h_1 *_H h_2).$$

Then the group $G \times H$ with operation $*$ is called the *direct product of G and H*. If G and H are abelian we sometimes write $G \oplus H$ instead of $G \times H$.

Example 8.5 (Direct product $\mathbb{Z}_2 \oplus \mathbb{Z}_2$)

The group $\mathbb{Z}_2$ is the group with elements $\{0, 1\}$ and operation $+$, as defined in the table below.

$+$	0	1
0	0	1
1	1	0

The group $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ therefore consists of the set of elements $\{(0,0), (0,1), (1,0), (1,1)\}$ with operation $*$ as defined in the table below

$*$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,1)$	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
$(1,0)$	$(1,0)$	$(1,1)$	$(0,0)$	$(0,1)$
$(1,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$

By writing $e = (0,0), a = (0,1), b = (1,0), c = (1,1)$ we get the multiplication table for the Klein 4 group, so $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ is the Klein 4 group (so in particular, it is not cyclic).

Exercise 8.6

The group $\mathbb{Z}_2$ is the group with elements $\{0, 1\}$ and operation $+$, as defined in the table below.

$+$	0	1
0	0	1
1	1	0

The group $\mathbb{Z}_3$ is the group with elements $\{0, 1, 2\}$ and operation $+$, as defined in the table below.

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

Write down the elements of the group $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ and its table. Show that the element $(1, 1)$ generates $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ and so $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is a cyclic group.

Exercise 8.7

Find the following products of elements in $S_3 \times \mathbb{Z}_3$:

(i) $\left(\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, 2\right)\left(\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, 1\right)$,

(ii) $\left(\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, 0\right)\left(\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, 1\right)$,

(iii) $\left(\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, 1\right)\left(\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, 1\right)$.

Exercise 8.8

(a) Find the cyclic subgroup $< (2,6) >$ of $\mathbb{Z}_4 \oplus \mathbb{Z}_{12}$ and hence find the order of $(2,6)$ in $\mathbb{Z}_4 \oplus \mathbb{Z}_{12}$.

(b) Find the cyclic subgroup $< (2,3) >$ of $\mathbb{Z}_4 \oplus \mathbb{Z}_{15}$ and hence find the order of $(2,3)$ in $\mathbb{Z}_4 \oplus \mathbb{Z}_{15}$.

(c) Find the cyclic subgroup $< (8,10) >$ of $\mathbb{Z}_{12} \oplus \mathbb{Z}_{18}$ and hence find the order of $(8,10)$ in $\mathbb{Z}_{12} \oplus \mathbb{Z}_{18}$.

Exercise 8.9

Find the order of the element

$$\left(a, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}\right)$$

in the group $V \times S_3$ where V is the Klein 4-group and S_3 is the symmetric group.

Exercise 8.10

Prove that a direct product $G \times H$ of groups G, H is abelian if and only if G, H are both abelian.

Exercise 8.11

Let $G = \mathbb{Z}_p \times \mathbb{Z}_q$ where p and q are prime. Show that every proper subgroup of G is cyclic, but that G itself is not necessarily cyclic.

Section 9. Homomorphisms and Isomorphisms

In this section we will consider maps between groups. The important words here will be *homomorphism* and *isomorphism,* which come from Greek words. The 'morphism' part comes from 'morphe' meaning 'form'. The distinction between 'iso' and 'homo' is more subtle. I will translate 'iso' as 'equal' and 'homo' as 'like' (a native Greek speaker may want to argue this point with me). So we are talking about 'like-forms' (for homomorphism) versus 'equal forms' (for isomorphism). We will make all this precise mathematically, but isomorphic groups will be groups that have equal forms (if not actually being equal – think about doing arithmetic in English, and then doing the same thing in French) and if there is a homomorphism from one group G to a group H then H has a 'like form' of G, i.e. some of the behaviour of G is preserved in H, but not necessarily all (we can think of H being a shadow of G).

Definition 9.1 (Homomorphism and Isomorphism)

Let G, H be groups. A map $\phi : G \to H$ is a *homomorphism* if $\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$. If, in addition, ϕ is one-one and onto then ϕ is an *isomorphism*. If there is an isomorphism $\phi : G \to H$ then we say that G and H are *isomorphic,* written $G \cong H$.

Example 9.2

(a) Consider the group G consisting of real numbers under addition, and consider the group H consisting of the set $\{e^{i\theta} \mid \theta \in \mathbb{R}\}$ under multiplication of complex numbers. Let $\phi : G \to H$ be given by $\phi(x) = e^{ix}$. Then

$$\phi(x+y) = e^{i(x+y)} = e^{ix}e^{iy} = \phi(x)\phi(y)$$

so ϕ is a homomorphism. But $\phi(x+2\pi) = e^{i(x+2\pi)} = e^{ix} = \phi(x)$ so ϕ is not one-one, so ϕ is not an isomorphism.

(b) Let G be the group of 2×2 matrices with real entries and non-zero determinant under matrix multiplication, and let H be the group of non-zero real numbers under multiplication. Let $\phi : G \to H$ be given by $\phi(A) = \det(A)$ (the determinant of A) for all $A \in G$. Then

$$\phi(AB) = \det(AB) = \det(A)\det(B) = \phi(A)\phi(B)$$

so ϕ is a homomorphism. It is not an isomorphism however, since (for example) $\phi(-I) = 1 = \phi(I)$ (where I is the 2×2 identity matrix).

(c) Let G be any group, and H be the *trivial group*, i.e. the group consisting of a single element e, say. Let $\phi : G \to H$ be given by $\phi(g) = e$ for all $g \in G$. Then $\phi(g_1g_2) = e = ee = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$ so ϕ is a homomorphism.The map ϕ is an isomorphism if and only if $|G| = 1$ (as this is the only way it can be one-one), i.e. G is the trivial group.

Example (Example 9.2 continued)

(d) Let $G = \mathbb{Z}$, the group of integers under addition and $H = 2\mathbb{Z}$, the group of even integers under addition. Let $\phi : G \to H$ be given by $\phi(x) = 2x$ for all $x \in G$. Then $\phi(x + y) = 2(x + y) = 2x + 2y = \phi(x) + \phi(y)$, so ϕ is a homomorphism. If $\phi(x) = \phi(y)$ then $2x = 2y$, i.e. $x = y$, so ϕ is one-one. For any $h \in H$ we have $h = 2t$ (for some $t \in \mathbb{Z}$) so $h = 2t = \phi(t) = \phi(g)$ where $g = t \in \mathbb{Z}$, so ϕ is onto. Therefore ϕ is an isomorphism, i.e. $\mathbb{Z} \cong 2\mathbb{Z}$. This should be surprising first time you see it. Certainly we do not have $\mathbb{Z} = 2\mathbb{Z}$, as $\mathbb{Z}$ contains odd numbers and $2\mathbb{Z}$ does not. So this illustrates the difference between equality and isomorphism. Isomorphism is about the algebraic structure being preserved, rather than the more restrictive consideration of things being identically equal. (Again compare doing arithmetic in English and in French.)

Example (Example 9.2 continued)

(e) Let $G = \mathbb{Z}_3$ and let $H = \{1, e^{2\pi i/3}, e^{4\pi i/3}\}$ under multiplication of complex numbers. Let $\phi : G \to H$ be given by $\phi(0) = 1$, $\phi(1) = e^{2\pi i/3}$, $\phi(2) = e^{4\pi i/3}$. Then ϕ is one-one and onto, moreover

$$\begin{aligned}
\phi(0+0) &= \phi(0) = 1 = 1 \cdot 1 = \phi(0) \cdot \phi(0) \\
\phi(0+1) &= \phi(1) = e^{2\pi i/3} = 1 \cdot e^{2\pi i/3} = \phi(0) \cdot \phi(1) \\
\phi(0+2) &= \phi(2) = e^{4\pi i/3} = 1 \cdot e^{4\pi i/3} = \phi(0) \cdot \phi(2) \\
\phi(1+1) &= \phi(2) = e^{4\pi i/3} = e^{2\pi i/3} \cdot e^{2\pi i/3} = \phi(1) \cdot \phi(1) \\
\phi(1+2) &= \phi(0) = 1 = e^{2\pi i/3} \cdot e^{4\pi i/3} = \phi(1) \cdot \phi(2) \\
\phi(2+2) &= \phi(1) = e^{2\pi i/3} = e^{4\pi i/3} \cdot e^{4\pi i/3} = \phi(2) \cdot \phi(2)
\end{aligned}$$

so ϕ is a homomorphism. Thus ϕ is an isomorphism, so $G \cong H$.

(f) Let G, H be groups. Let $\phi : G \times H \to G$ be given by $\phi((g,h)) = g$ for all $(g,h) \in G \times H$. Then, given $(g_1, h_1), (g_2, h_2) \in G \times H$ we have

$$\phi((g_1,h_1)(g_2,h_2)) = \phi((g_1g_2, h_1h_2)) = g_1g_2 = \phi((g_1,h_1))\phi((g_2,h_2))$$

so ϕ is a homomorphism. If H is non-trivial then ϕ is not one-one (because $\phi(g, e_H) = \phi(g,h)$ for any $h \in H$, $h \neq e_H$, where e_H is the identity of H), so ϕ is not an isomorphism. If H is the trivial group then ϕ is one-one and onto so ϕ is an isomorphism.

We now give some properties of homomorphisms.

Lemma 9.3

Let G, H be groups with identity elements e_G, e_H and let $\phi : G \to H$ be a homomorphism. Then

(a) $\phi(e_G) = e_H$,

(b) $\phi(g^{-1}) = \phi(g)^{-1}$ *(for all $g \in G$)*,

(c) $\phi(G) = \{\phi(g) \mid g \in G\}$ *is a subgroup of H.*

Proof.

(a) Let $g \in G$. Then

$$\phi(g) = \phi(e_G g) = \phi(e_G)\phi(g)$$

and

$$\phi(g) = e_H \phi(g)$$

so $e_H\phi(g) = \phi(e_G)\phi(g)$ and left cancelling gives $e_H = \phi(e_G)$.

(b) We have $e_H = \phi(e_G) = \phi(gg^{-1}) = \phi(g)\phi(g^{-1})$ so multiplying (on the left) by $\phi(g)^{-1}$ we get $\phi(g)^{-1}e_H = \phi(g^{-1})$ i.e. $\phi(g)^{-1} = \phi(g^{-1})$.

(c) We show that $\phi(G)$ is a group. Multiplication in $\phi(G)$ is associative (see Exercise 9.10). Let $\phi(g_1), \phi(g_2) \in \phi(G)$. By (a) we have $\phi(g_1)\phi(g_2) = \phi(g_1 g_2) \in \phi(G)$ so G is closed under multiplication. Then $\phi(e_G) \in \phi(G)$ is the identity of $\phi(G)$. Given $\phi(g) \in \phi(G)$ we have $\phi(g^{-1}) \in \phi(G)$ and $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(e_G)$ and $\phi(g^{-1})\phi(g) = \phi(g^{-1}g) = \phi(e_G)$ so $\phi(g^{-1})$ is the inverse of $\phi(g)$.

□

Example 9.4

Let $G = \mathbb{R}$ be the group of real numbers under addition and let $H = \mathbb{C}^*$, the group of non-zero complex numbers under multiplication. Let $\phi : G \to H$ be given by $\phi(x) = e^{ix}$. Lemma 9.3(c) implies that $\phi(G) = \{e^{ix} \mid x \in \mathbb{R}\} = \{e^{ix} \mid 0 \leq x < 2\pi\}$ is a subgroup of $\mathbb{C}^*$. We originally showed this in Example 3.3(i) by checking the four properties of a group, but now we have a more direct proof of this fact.

We now give some properties of groups that are preserved under isomorphism.

Lemma 9.5

Let $G \cong H$.

(a) *if $|G| = n$ ($n \in \mathbb{N}$) then $|H| = n$;*

(b) *if $|G| = \infty$ then $|H| = \infty$;*

(c) *if G is abelian then H is abelian;*

(d) *if G is cyclic then H is cyclic;*

Proof.

(a) This is because any isomorphism $\phi : G \to H$ is one-one.

(b) This is because any isomorphism $\phi : G \to H$ is one-one.

(c) Since $G \cong H$ there exists some isomorphism $\phi : G \to H$. Suppose G is abelian and let h_1, h_2 in H. We must show that $h_1 h_2 = h_2 h_1$. Now $h_1 = \phi(g_1), h_2 = \phi(g_2)$ for some g_1, g_2 in G and

$$h_1 h_2 = \phi(g_1)\phi(g_2) = \phi(g_1 g_2) = \phi(g_2 g_1) = \phi(g_2)\phi(g_1) = h_2 h_1$$

as required.

(d) Exercise 9.11.

□

We can use such properties to show that certain pairs of groups are not isomorphic.

Example 9.6

(a) $\mathbb{Z} \ncong \mathbb{Z}_n$ (as their orders are different);

(b) $S_4 \ncong D_8$ (as $|S_4| = 4! = 24$ and $|D_8| = 8$);

(c) the group of invertible 2×2 matrices with real entries under multiplication is not isomorphic to the group of real numbers under addition, as the former is not abelian but the latter is.

Exercise 9.7

(a) Let G be the group $\mathbb{R}^*$ of non-zero real numbers under multiplication, and let H be the group $\mathbb{R}^*$ of non-zero real numbers under multiplication. Show that the map $f : G \to H$ given by $f(x) = |x|$ (the absolute value of x) is a homomorphism. Is f an isomorphism? (Justify your answer.)

(b) Let G be the group of all polynomials in a variable x with real coefficients under the operation of polynomial addition. (Check you understand why this is a group.) Let $H = G$. Let $\phi : G \to H$ be given by $\phi(f) = f'$ (i.e. ϕ takes f to its derivative). Show that ϕ is a homomorphism. Is ϕ an isomorphism? (Justify your answer.)

(c) Let $G = \mathbb{Z}_4$, $H = \{1, i, -1, -i\}$ under multiplication of complex numbers. Show that $G \cong H$.

Exercise 9.8

Let G be the group of real numbers under addition and let H be the group of positive real numbers under multiplication. By using the exponential function, show that $G \cong H$.

Exercise 9.9

Determine which of the following are homomorphisms and which are isomorphisms.

(i) $\phi : \mathbb{Z} \to \mathbb{R}$ given by $\phi(n) = n$;

(ii) $\phi : 2\mathbb{Z} \to \mathbb{Z}$ given by $\phi(n) = n$;

(iii) $\phi : 2\mathbb{Z} \to \mathbb{Z}$ given by $\phi(n) = n/2$;

(iv) $\phi : \mathbb{R} \to \mathbb{Z}$ given by $\phi(x) =$ the greatest integer $\leq x$;

(v) $\phi : G \to G$ given by $\phi(g) = g^{-1}$, where G is any abelian group;

(vi) $\phi : G \to G$ given by $\phi(g) = g^{-1}$, where G is any non-abelian group;

(vii) $\phi : G \times H \to H \times G$ given by $\phi((g, h)) = (h, g)$ for all $g \in G, h \in H$, where G, H are any groups.

Exercise 9.10

Let G be a group, let H be a set with a closed binary operation and let $\phi : G \to H$ be a homomorphism. Show that for $g_1, g_2, g_3 \in G$ we have

$$(\phi(g_1)\phi(g_2))\,\phi(g_3) = \phi(g_1)\,(\phi(g_2)\phi(g_3))\,.$$

(That is, show that multiplication in $\phi(G)$ is associative.)

Exercise 9.11

Let $G \cong H$. Show that if G is cyclic then H is cyclic.

Section 10. Rings and fields.

So far we have been concerned with algebraic objects with a single binary operation. Our experience in other parts of mathematics shows that it often makes sense to deal with algebraic operations with two binary operations, addition and multiplication. We now move on to such objects, which are called Rings.

Definition 10.1 (Ring)

A *ring* R is a nonempty set with two closed binary operations, addition and multiplication, which we denote $a+b$ and ab respectively, such that the following hold:

(i) $a+b=b+a$ for all $a,b\in R$,

(ii) $(a+b)+c=a+(b+c)$ for all $a,b,c\in R$,

(iii) there exists some element $0\in R$ such that $a+0=a$ for all $a\in R$ (we call 0 the *additive identity* of R),

(iv) for each $a\in R$ there exists some element $-a\in R$ such that $a+(-a)=0$,

(v) $a(bc)=(ab)c$ for all $a,b,c\in R$,

(vi) $a(b+c)=ab+ac$ and $(b+c)a=ba+ca$, for all $a,b,c\in R$.

Note that conditions (i),(ii),(iii),(iv) say that R is an abelian group under the operation $+$. There is **no requirement** that R is a group under the multiplicative operation, however. Condition (v) says that the multiplication operation is associative. The conditions in Condition (vi) are called the *left distributive law* (for the first one) and the *right distributive law* (for the second one).

Example 10.2

(a) The set of integers $\mathbb{Z}$ together with the usual addition and multiplication of integers forms a ring.

(b) The set $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ together with the addition operation that defines $a + b$ to be the remainder when the integer $a + b$ is divided by n (*addition modulo n*) and defines ab to be the remainder when the integer ab is divided by n (*multiplication modulo n*) is a ring.

(c) The set of all 2×2 matrices with integer entries forms a ring under addition and multiplication of matrices.

(d) The set $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ forms a ring under the usual addition and multiplication of integers.

(e) We write $\mathbb{Z}[x]$ to denote all polynomials in the variable x with integer coefficients. So, for example, $3 + x - x^5 \in \mathbb{Z}[x]$, but $x^2/2 \notin \mathbb{Z}[x]$. The set $\mathbb{Z}[x]$ together with the usual addition and multiplication operations for polynomials forms a ring.

(f) The set of complex numbers together with the usual addition and multiplication of complex numbers forms a ring.

(g) The set R consisting only of the additive identity 0 forms a ring, called the *trivial ring.*

Definition 10.3 (More terminology about rings)

(a) If the multiplication operation is commutative (i.e. $ab = ba$ for all $a, b \in R$) then we say that R is a *commutative ring*.

(b) If there is some element $1 \in R$ such that $1a = a1 = a$ for all $a \in R$ then this element is called a *unity* and R is said to be a *ring with unity.*

Example 10.4

Consider Example 10.2. Part (a) is a commutative ring with unity (the unity being the integer 1). Part (b) is a commutative ring with unity (the unity being the element 1). Part (c) is a noncommutative ring with unity (the unity being the 2×2 identity matrix). Part (d) is a commutative ring without unity. Part (e) is a commutative ring with unity (the unity being the polynomial 1).Part (f) is a commutative ring with unity (unity being the complex number 1). Part (g) is a commutative ring with unity (where the unity is the ring element 0).

A comment on notation

Sometimes when working in a ring we may need to add something to itself many times. For example in the ring of 2×2 matrices with integer entries we may need to calculate

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

We will write this as

$$4 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

This seems reasonable, but remember that 4 is not a 2×2 matrix, so this does not refer to multiplication in the ring. This is an unfortunate clash of notation, but we have to live with it. That is, in general for an element a in a ring R, instead of writing

$$\underbrace{a + \ldots + a}_{n}$$

where $n \geq 1$, we will write na, and if $n \leq -1$ we will write na to mean

$$\underbrace{(-a) + \ldots + (-a)}_{|n|}.$$

We now derive some rules for multiplying in rings. The statement looks completely obvious if you only think about rings of real numbers, integers, or matrices, but remember that we are now dealing with much more general objects so we need to prove that these 'obvious' properties hold. Note that rings have two operations, addition and multiplication, denoted $a + b$ and ab; there is no subtraction operation. However, each element $a \in R$ has an additive inverse, denoted $(-a)$. We can therefore write $a - b$ to mean $a + (-b)$.

Theorem 10.5 (Multiplication rules for rings)

Let a, b, c be elements of a ring R with additive identity 0. Then

(a) $a0 = 0a = 0$,

(b) $a(-b) = -(ab)$ *and* $(-a)b = -(ab)$,

(c) $(-a)(-b) = ab$,

(d) $a(b - c) = ab - ac$ *and* $(b - c)a = ba - ca$.

If R has a unity element 1 then

(e) $(-1)a = -a$,

(f) $(-1)(-1) = 1$.

Proof.

(a) $0 + a0 = a0 = a(0 + 0) = a0 + a0$. Using the cancellation property of groups (Lemma 3.12) gives $0 = a0$. Similarly we get $0a = 0$.

(b) By definition $-(ab)$ is the element which when added to ab gives zero. Thus to show $a(-b) = -(ab)$ we must show that $a(-b) + ab = 0$. This holds because

$$a(-b) + ab = a(-b + b) = a0 = 0.$$

Similarly

$$(-a)b + ab = (-a + a)b = 0b = 0$$

shows that $(-a)b = -(ab)$.

(c) The second identity of part (b) gives

$$\begin{aligned}(-a)(-b) &= -(a(-b)) \\ &= -(-(ab)) \\ &= -c\end{aligned}$$

where $c = -(ab)$. Now $-c$ is the additive inverse of c, but $-c = (-a)(-b)$ so $(-a)(-b)$ is the additive inverse of c. But ab is the additive inverse of c so $(-a)(-b) = ab$, as required.

Proof of Theorem 10.5 continued.

(d)

$$\begin{aligned} a(b-c) &= a(b+(-c)) \\ &= ab + a(-c) \quad \text{by the left distributive law} \\ &= ab + (-(ac)) \quad \text{by (b)} \\ &= ab - ac. \end{aligned}$$

The second identity is similar (see Exercise 10.15).

(e) Putting $a = 1$, $b = a$ in the second identity of part (b) gives $(-1)a = -(1a)$, i.e. $(-1)a = -a$.

(f) Putting $a = 1, b = 1$ in (c) gives $(-1)(-1) = 1 \cdot 1 = 1$.

□

Theorem 10.6 (Direct product of rings)

Let R, S be rings. Define addition and multiplication on the cartesian product $R \times S$ by

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2),$$
$$(r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2).$$

Then $R \times S$ together with these binary operations is a ring.

Proof.

Exercise 10.17. □

Definition 10.7 (Direct product of rings)

Let R, S be rings. Define addition and multiplication on the cartesian product $R \times S$ by

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2),$$
$$(r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2).$$

Then the ring $R \times S$ with these binary operations is called the *direct product of R and S*.

Example 10.8

(a) The ring $\mathbb{Z}[x] \times \mathbb{C}$ is a commutative ring with unity $(1, 1)$ and in this ring we have we have

$$(x^2 + x + 1, 2 + i) + (x^3 - 1, -1 - i) = (x^3 + x^2 + x, 1),$$
$$(x^2 + x + 1, 2 + i)(x^3 - 1, -1 - i) = (x^5 + x^4 + x^3 - x^2 - x - 1, -1 - 3i).$$

(b) Let R be the ring of 2×2 matrices with integer entries, and let S be the ring $\mathbb{Z}_6$. Then $R \times S$ is a ring with unity $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, 1\right)$, but it is not commutative, since R is not commutative. In this ring we have

$$\left(\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}, 5\right) + \left(\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, 3\right) = \left(\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, 2\right),$$
$$\left(\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}, 5\right)\left(\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, 3\right) = \left(\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}, 3\right).$$

Just as we have the concept of subset and subgroup, we also have the concept of subring.

Definition 10.9 (Subring)

Let R be a ring and let S be a subset of R. Then S is a *subring of* R if S is a ring under the same addition and multiplication operations of R. If S is a subring of R then we write $S \leq R$ and write $S < R$ if we also have that $S \neq R$.

Example 10.10

(a) For a ring R the trivial ring $\{0\}$ is a subring of R.

(b) For a ring R the ring R a subring of R.

(c) $S = \{0, 2, 4\}$ is a subring of $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$, but $T = \{0, 1, 2\}$ is not (for example $1 + 2 \notin T$).

(d) The set of *Gaussian integers* $\mathbb{Z}[i] = \{a + ib \mid a, b \in \mathbb{Z}\}$ is a subring of the ring of complex numbers $\mathbb{C}$, as can be seen by checking the properties of rings (i)–(vi) from Definition 10.1.

(e) The set S of diagonal matrices with integer entries

$$\left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \;\middle|\; a, b \in \mathbb{Z} \right\}$$

is a subring of the ring R of all 2×2 matrices with integer entries, as can be seen by checking the properties of rings (i)–(vi) from Definition 10.1.

Theorem 10.11 (Subring test)

A nonempty subset S of a ring R is a subring of R if and only if for all $a, b \in S$ we have $ab \in S$ and $a - b \in S$.

Proof.

Suppose first that S is a subring of R. Then S is a ring. Then for $a, b \in S$ we have $ab \in S$ and $(-b) \in S$ so $a + (-b) \in S$, i.e. $a - b \in S$. That is, the stated conditions hold.
Suppose then that S is a non-empty subset of R such that for all $a, b \in S$ we have $ab \in S$ and $a - b \in S$. We must show that S is a ring. By assumption we have that S is closed under the multiplication operation. Since conditions (i),(ii),(v),(vi) of Definition 10.1 hold for all elements of R, they also hold for all elements of S, so we only need to check conditions (iii) (existence of additive identity),(iv) (existence of additive inverses), and also closure under addition. Putting $a = b$ in the second condition gives that $a - a \in S$, i.e. $0 \in S$, so (iii) is satisfied. Putting $a = 0$ in the first condition gives $0 - b \in S$, i.e. $-b \in S$, so (iv) is satisfied. Thus $a - (-b) \in S$, i.e. $a + b \in S$, so S is closed under the addition operation. □

Example 10.12

Consider parts (c),(d),(e) of Example 10.10.

(c) $S = \{0, 2, 4\}$, $R = \mathbb{Z}_6$. Then the product of any two elements of S (mod 6) is an element of S, and the difference of any two elements of S (mod 6) is an element of S. Therefore S is a subring of R. $T = \{0, 1, 2\}$, but here $0 - 1 = 5 \notin T$, so T is not a subring of R.

(d) $S = \mathbb{Z}[i]$, $R = \mathbb{C}$. Let $a + ib, c + id \in \mathbb{Z}[i]$. Then
$(a + ib)(c + id) = (ac - bd) + (ac + bd)i \in \mathbb{Z}[i]$ and
$(a + ib) - (c + id) = (a - c) + i(b - d) \in \mathbb{Z}[i]$ so $\mathbb{Z}[i]$ is a subring of $\mathbb{C}$.

(e) S is the set of all 2×2 diagonal matrices with integer entries, R is the ring of all 2×2 matrices with integer entries. Let $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \in S$. Then

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} ac & 0 \\ 0 & bd \end{pmatrix} \in S,$$

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} - \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} a - c & 0 \\ 0 & b - d \end{pmatrix} \in S.$$

So S is a subring of R.

As with groups, we also have the concepts of homomorphism and isomorphism.

Definition 10.13 (Homomorphism and Isomorphism for rings)

Let R and S be rings. A map $\phi : R \to S$ is a *homomorphism* if for all $a, b \in R$ the following hold:

(i) $\phi(a + b) = \phi(a) + \phi(b)$,

(ii) $\phi(ab) = \phi(a)\phi(b)$.

If, in addition, ϕ is one-one and onto then ϕ is an *isomorphism.* If there exists an isomorphism $\phi : R \to S$ then R and S are *isomorphic* and we write $R \cong S$.

Example 10.14

(a) Let $R = \mathbb{Z}$, $S = 2\mathbb{Z}$ be rings with the usual addition and multiplication operations. The map $\phi : R \to S$ be given by $\phi(x) = 2x$ for all $x \in R$. Now $\phi(xy) = 2xy$ but $\phi(x)\phi(y) = (2x)(2y) = 4xy \neq \phi(xy)$ (for $x \neq 0, y \neq 0$) so ϕ is not a ring homomorphism. Contrast this with Example 9.2(d) where the map was an isomorphism of groups.

(b) Let $R = \mathbb{Z}[x]$, the ring of polynomials in x with integer coefficients and let $S = \mathbb{Z}$ be the ring of integers. Let $\phi : R \to S$ be given by $\phi(f) = f(0)$ for all $f \in \mathbb{Z}[x]$. (In other words ϕ maps f to the constant term of f.) Then writing

$$f(x) = \sum_{i=0}^{m} a_i x^i \in R,$$

$$g(x) = \sum_{j=0}^{n} b_j x^j \in R,$$

for some $m, n \geq 0$ we have that the constant term of $(f + g)(x)$ is $a_0 + b_0$ so $(f + g)(0) = a_0 + b_0$ and so

$$\phi(f) + \phi(g) = f(0) + g(0) = a_0 + b_0 = (f + g)(0) = \phi(f + g).$$

Similarly, the constant term of $(fg)(x)$ is $a_0 b_0$ so $(fg)(0) = a_0 b_0$ and so

$$\phi(f)\phi(g) = f(0)g(0) = a_0 b_0 = fg(0) = \phi(fg).$$

Thus ϕ is a (ring) homomorphism.

Example (Example 10.14 continued)

(c) Let $R = \mathbb{Z}[i] = \{a + ib \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers, and let $S = \mathbb{Z}$ be the ring of integers. Let $\phi : R \to S$ be given by $\phi(a + ib) = a$. Then (for example)

$$\phi(1 + i)\phi(1 - i) = 1 \cdot 1 = 1$$

and

$$\phi((1 + i)(1 - i)) = \phi(2) = 2$$

so ϕ is not a homomorphism.

Example (Example 10.14 continued)

(d) The set R of matrices of the form

$$\left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{Z} \right\}$$

forms a ring under addition and multiplication of matrices. Let $\phi : R \to \mathbb{Z}$ be given by

$$\phi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right) = a.$$

Then

$$\begin{aligned} \phi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} d & e \\ 0 & f \end{pmatrix}\right) &= \phi\left(\begin{pmatrix} ad & ae+bf \\ 0 & cf \end{pmatrix}\right) = ad \\ &= \phi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right)\phi\left(\begin{pmatrix} d & e \\ 0 & f \end{pmatrix}\right) \end{aligned}$$

and

$$\begin{aligned} \phi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} + \begin{pmatrix} d & e \\ 0 & f \end{pmatrix}\right) &= \phi\left(\begin{pmatrix} a+d & b+e \\ 0 & c+f \end{pmatrix}\right) = a+d \\ &= \phi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right) + \phi\left(\begin{pmatrix} d & e \\ 0 & f \end{pmatrix}\right) \end{aligned}$$

so ϕ is a homomorphism.

Exercise 10.15

Show that if a, b, c are elements of a ring R we have $(b - c)a = ba - ca$.

Exercise 10.16

(i) Calculate $15 + 12$ and $(15)(12)$ in the ring $\mathbb{Z}_{24}$.

(ii) Calculate $14 + 9$ and $(11)(10)$ in the ring $\mathbb{Z}_{15}$.

Exercise 10.17

Let R, S be rings. Define addition and multiplication on the cartesian product $R \times S$ by

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2),$$
$$(r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2).$$

Show that $R \times S$ together with these binary operations is a ring.

Exercise 10.18

Write down all elements of the ring $R = \mathbb{Z}_3 \oplus \mathbb{Z}_5$. Does R have a unity (if so, what is it)?

Exercise 10.19

Let R be the ring of 2×2 matrices with real entries, and let S be the following subset of R:

$$S = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \,\middle|\, a, b, \in \mathbb{R} \right\}.$$

(a) Show that S is a subring of R.

(b) Show that the map $\phi : \mathbb{C} \to S$ given by

$$\phi(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

(for $a, b \in \mathbb{R}$) is a ring isomorphism.

Exercise 10.20

An element a of a ring R is called an *idempotent* if $a^2 = a$.

(a) Find all the idempotents in the rings (i) $\mathbb{Z}_4$, (ii) $\mathbb{Z}_6$, (iii) $\mathbb{Z}_4 \oplus \mathbb{Z}_6$.

(b) Show that in a commutative ring R, the product of any two idempotents is an idempotent.

Exercise 10.21

Let R be a ring with additive identity 0, and let S, T be subrings of R. Show that the intersection $S \cap T$ is a subring of R.

Exercise 10.22

Let R be a ring. Show that R is a commutative ring if and only if $a^2 - b^2 = (a - b)(a + b)$ for all $a, b \in R$.

Exercise 10.23

Let $\mathbb{Z}_n[x]$ denote the ring of polynomials in x whose coefficients are in the ring $\mathbb{Z}_n$, where the addition and multiplication operations in $\mathbb{Z}_n[x]$ are the usual addition and multiplication operations, except that the coefficients are reduced mod n. (It follows from the ring axioms that $\mathbb{Z}_n[x]$ is a ring.)
Let

$$f(x) = 5x^3 + 3x^2 + 2x + 4$$
$$g(x) = 6x^2 + 5x + 1$$

be polynomials in $\mathbb{Z}_7[x]$. Find $f(x) + g(x)$ and $f(x)g(x)$ in $\mathbb{Z}_7[x]$.

Section 11. Multiplicative behaviour and Fields.

First we observe that a ring can have at most one unity element.

Lemma 11.1

Let R be a ring with unity 1. If there exists $e \in R$ such that $ea = ae = a$ for all $a \in R$ then $e = 1$.

Proof.

Since 1 is a unity we have $1e = e$, since e is a unity we have $1e = 1$. Thus $e = 1$. □

Definition 11.2 (Unit)

Let R be a ring with unity 1. Given $a \in R$, a *multiplicative inverse of a* is an element $b \in R$ such that $ab = ba = 1$. We normally write a^{-1} for this element b. An element $u \in R$ is called a *unit* if it has a multiplicative inverse in R.

(Be careful not to confuse the words *unit* and *unity.*) As in Lemma 3.16, if an element $a \in R$ has a multiplicative inverse, then that inverse is unique. For a given ring with unity, some elements might have inverses, while others do not.

Example 11.3

Consider the ring $\mathbb{Z}_{14} = \{0, 1, 2, \ldots, 13\}$. The unity is the element 1. Now $(3)(5) = 1$ and $(5)(3) = 1$ in $\mathbb{Z}_{14}$ so 3 has a multiplicative inverse, namely 5. By the same argument 5 has a multiplicative inverse, namely 3. Similarly $(9)(11) = 1$ and $(11)(9) = 1$ in $\mathbb{Z}_{14}$ so 9 and 11 are multiplicative inverses of each other. The element 13 is its own multiplicative inverse, as is 1. However, there is no element $a \in \mathbb{Z}_{14}$ such that $0a = a0 = 1$ so 0 has no multiplicative inverse. Further, there is no element $a \in \mathbb{Z}_{14}$ such that $2a = a2 = 1$ so 2 has no multiplicative inverse. In the same way $4, 6, 7, 8, 10, 12$ do not have multiplicative inverses. Thus the units are $1, 3, 5, 9, 11, 13$.

(In general, it can be shown that in the ring $\mathbb{Z}_n$, the units are those elements u such that the greatest common divisor $(u, n) = 1$.)

Example 11.4

In the ring of integers $\mathbb{Z}$ the unity element is the integer 1 and we have $1 \cdot 1 = 1$ and $(-1) \cdot (-1) = 1$ so 1 and -1 are units. If $n \neq \pm 1$ then there is no m such that $nm = 1$, so n is not a unit. Thus the units in $\mathbb{Z}$ are $1, -1$.

Definition 11.5 (Field)

A *field* is a commutative ring with unity in which every non-zero element is a unit.

Example 11.6

(a) $\mathbb{Z}$ is not a field (as not all non-zero elements are units).

(b) Consider the ring $\mathbb{Q}$. Clearly $\mathbb{Q}$ is commutative and has unity $1 = 1/1$. Given a non-zero element $p/q \in \mathbb{Q}$, we have $(p/q)(q/p) = 1$ so p/q is a unit, and hence $\mathbb{Q}$ is a field.

(c) Consider the ring $\mathbb{C}$. Clearly $\mathbb{C}$ is commutative and has unity $1 = 1 + 0i$. Given a non-zero element $a + bi \in \mathbb{C}$, we have $(a + bi)\left(\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i\right) = 1$ so $a + bi$ is a unit, and hence $\mathbb{C}$ is a field.

(d) Consider the set $\mathbb{Z}_3[i] = \{a + bi \mid a, b \in \mathbb{Z}_3\}$ together with addition and multiplication, which are the usual addition and multiplication operations of $\mathbb{C}$, except that the coefficients are reduced mod 3. (For example, $(1 + i) + (2i) = 1$, $(1 + i)(2i) = 1 + 2i$.) Then $\mathbb{Z}_3[i]$ can be shown to be a ring (with nine elements). Clearly it is commutative, and has unity $1 = 1 + 0i$. Then $1.1 = 1$ so 1 is a unit, $2.2 = 1$ so 2 is a unit, $(i)(2i) = 1$ so $i, 2i$ are units, $(1 + i)(2 + i) = 1$ so $1 + i, 2 + i$ are units, $(1 + 2i)(2 + 2i) = 1$ so $1 + 2i, 2 + 2i$ are units. Thus every non-zero element is a unit, so $\mathbb{Z}_3[i]$ is a field.

Definition 11.7 (Characteristic)

The *characteristic* of a ring R, denoted $\text{char}(R)$, is the least positive integer n such that $na = 0$ for all $a \in R$. If no such integer exists we say that R has characteristic 0.

Example 11.8

(a) In the ring $\mathbb{Z}_{12}$ we have that $12x = 0$ for all $x \in \mathbb{Z}_{12}$. Moreover, 12 is the least such positive integer since $n(1) \neq 0$ for $n < 12$. Thus $\text{char}(\mathbb{Z}_{12}) = 12$. In the subring $S = \{0, 4, 8\}$ we have $3.a = 0$ for all $a \in S$, and 3 is the least such positive integer, since $1(4) \neq 0$ and $2(4) \neq 0$. Thus $\text{char}(S) = 3$.

(b) In the ring $\mathbb{Z}$ there is no positive integer n such that $nx = 0$ for all $x \in \mathbb{Z}$ so $\text{char}(\mathbb{Z}) = 0$ (i.e. $\mathbb{Z}$ has characteristic 0).

(c) The set $\mathbb{Z}_2[x]$ of all polynomials with coefficients in $\mathbb{Z}_2 = \{0, 1\}$ together with addition and multiplication operations that are the ordinary addition and multiplication operations for polynomials, except that coefficients are considered mod 2, form a ring. (For example $(1 + x + x^3) + (1 + x) = 2 + 2x + x^3 = x^3$,
$(1 + x + x^3)(1 + x) = 1 + 2x + x^3 + x^2 + x^4 = 1 + x^2 + x^3 + x^4$.) For $p(x) \in \mathbb{Z}_2[x]$ we have $2p(x) = 0$, and moreover, there exist polynomials $p(x)$ such that $1p(x) \neq 0$, so $\text{char}(\mathbb{Z}_2[x]) = 2$. Thus infinite rings can have finite characteristic.

The following theorem says that, for rings with unity, to calculate the characteristic of the ring, we only need to consider the unity element.

Theorem 11.9

Let R be a ring with unity 1. If 1 has infinite order under addition then $char(R) = 0$. If 1 has order $n < \infty$ under addition, then $char(R) = n$.

Proof.

If 1 has infinite additive order, then the cyclic subgroup $< 1 >$ of the ring R under the addition operation, is infinite. Thus there is no finite positive integer n such that $n(1) = 0$. Thus $\text{char}(R) = 0$.
Suppose then that 1 has additive order $n < \infty$. Then the cyclic subgroup $< 1 >$ of the ring R under addition has n elements, namely

$$0, 1, 1+1, 1+1+1, \ldots, \underbrace{1+1+\ldots+1}_{n-1}.$$

Therefore $n(1)$ is one of these elements. If $n(1) = \underbrace{1+1+\ldots+1}_{t}$ $(1 \leq t \leq n-1)$ then $(n-t)(1) = 0$ so 1 has order at most $(n-t)$, a contradiction. Therefore $n(1) = 0$ and $m(1) \neq 0$ for $m < n$. Thus $\text{char}(R) \geq n$.

Proof of Theorem 11.9 continued.

Now for any $a \in R$ we have

$$\begin{aligned} n(a) &= \underbrace{a + a + \ldots + a}_{n} \\ &= \underbrace{1(a) + 1(a) + \ldots + 1(a)}_{n} \\ &= (\underbrace{1 + 1 + \ldots + 1}_{n})(a) \\ &= (n(1))(a) \\ &= 0. \end{aligned}$$

Thus $\text{char}(R) \leq n$ and so $\text{char}(R) = n$. □

Exercise 11.10

Find all the units in each of the following rings:

(a) $\mathbb{Z}_{12}$,

(b) $\mathbb{Z}_5$,

(c) $\mathbb{Z}_4$,

(d) $\mathbb{Z}_6$,

(e) $\mathbb{Z}_4 \oplus \mathbb{Z}_{12}$.

(f) $\mathbb{R}$,

(g) the ring of 2×2 matrices with real entries.

In each case say if the ring is a field.

Exercise 11.11

Show that the set $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$, under the usual addition and multiplication operations for real numbers, is a field. (Note that first you must show that it is a ring.)

Exercise 11.12

(a) Give an example of a ring R with unity such that $a^n = 0$ in R for some $a \in R$, $n \geq 2$, where $a \neq 0$.

(b) Let R be a ring with unity which has an element $a \in R$ such that $a^n = 0$ for some $n \geq 2$. Show that $(1 - a)$ is a unit in R.

Exercise 11.13

Find the characteristic of the following rings:

(a) $\mathbb{Z} \oplus \mathbb{Z}$,

(b) $\mathbb{Z}_3 \oplus 3\mathbb{Z}$,

(c) $\mathbb{Z}_3 \oplus \mathbb{Z}_3$,

(d) $\mathbb{Z}_3 \oplus \mathbb{Z}_4$.

Exercise 11.14

(a) Expand and simplify $(a+2b)^3$ in a ring of characteristic 4.

(b) Expand and simplify $(a+b)^6$ in a ring of characteristic 3.

Section 12. Zero divisors and Integral domains.

Consider the ring $R = \mathbb{Z}_{12} = \{0, 1, \ldots, 11\}$ under addition and multiplication mod 12. We have

$$3 \cdot 4 = 0$$

in R. So what? Well, it means that there is a solution to the equation

$$xy = 0$$

with $x \neq 0$ and $y \neq 0$. This is strange, and very different to what happens when we deal with real numbers, where we are used to being able to write "if $xy = 0$ then $x = 0$ or $y = 0$".

Definition 12.1 (Zero-divisor)

Let R be a ring. If a, b are non-zero elements of R such that $ab = 0$ in R then a, b are *zero-divisors* in R.

Example 12.2

(a) In the ring $\mathbb{Z}_{12}$ the zero-divisors are $2, 3, 4, 6, 8, 9, 10$.

(b) In the ring of 2×2 matrices with integer entries we have

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

so these two matrices are zero divisors.

In fact, in the ring $\mathbb{Z}_n$ we can say precisely what the zero divisors are. For its proof, we will need Bézout's identity, which we state without proof. We say that an integer m *divides* an integer n if n is an integral multiple of m. We say that two integers m, n are *coprime* if their greatest common divisor (m, n) (or highest common factor) is equal to 1. (That is, if d divides m and d divides n then $d = 1$.)

Theorem 12.3 (Bézout's identity)

Let $m, n \in \mathbb{Z}$ be coprime integers. Then there exist integers $p, q \in \mathbb{Z}$ such that $pm + qn = 1$.

Corollary 12.4

Let $m, n \in \mathbb{Z}$ be coprime integers. If n divides ms for some $s \in \mathbb{Z}$ then n divides s.

Proof.

Since $(m, n) = 1$, Bézout's identity shows that there exist integers p, q such that $pm + qn = 1$. Then $pms + qns = s$. Now n divides qns and n divides ms, so it divides pms, thus n divides $pms + qns = s$, i.e. n divides s, as required. □

Theorem 12.5 (Zero divisors in $\mathbb{Z}_n$)

In the ring $\mathbb{Z}_n$ the zero divisors are precisely those non-zero elements $m \in \mathbb{Z}_n$ such that the greatest common divisor $(m, n) > 1$.

Proof.

Let $m \in \mathbb{Z}_n$, $m \neq 0$ such that $d = (m, n) > 1$. Then

$$m\left(\frac{n}{d}\right) = \left(\frac{m}{d}\right) n$$

which is a multiple of n, so when $m(n/d)$ is divided by n it leaves a remainder of zero, so $m(n/d) = 0$ in $\mathbb{Z}_n$ and since $n/d \neq 0$ in $\mathbb{Z}_n$ we conclude that m is a zero divisor.
Suppose then that $m \in \mathbb{Z}_n$ satisfies $m \neq 0$, $(m, n) = 1$, and m is a zero-divisor. Then $ms = 0$ in $\mathbb{Z}_n$ for some $s \in \mathbb{Z}_n$, $s \neq 0$. In particular, $s \in \{1, \ldots, n-1\}$. Thus ms leaves a remainder of zero when divided by n, i.e. n divides ms. By Corollary 12.4, since $(m, n) = 1$ we have that n divides s, a contradiction. □

Corollary 12.6

If p is prime then the ring $\mathbb{Z}_p$ has no zero divisors.

Proof.

By Theorem 12.5 we have that if $m \in \mathbb{Z}_p$ is a zero divisor, then $m \in \{1, \ldots, p-1\}$ and the greatest common divisor $(m, p) > 1$. But the divisors of p are 1 and p, so $(m, p) = p$, so p divides m, a contradiction. □

Example 12.7

We shall solve the quadratic equation $x^2 - 5x + 6 = 0$ in the ring $\mathbb{Z}_{12}$. We have $x^2 - 5x + 6 = (x-2)(x-3)$, so the equation is clearly satisfied if $(x-2) = 0$ or $(x-3) = 0$, i.e. for $x = 2$ or 3. However there may be other solutions. Since $(x-2)(x-3) = 0$ they are a pair of zero divisors that differ by 1. The zero divisors in $\mathbb{Z}_{12}$ are $2, 3, 4, 6, 8, 9, 10$. Putting $(x-3)$ equal to each of these in turn gives a solution if and only if $(x-3) = 3$ or $(x-3) = 8$ so $x = 6$ or 11. Thus all the solutions are $x = 2, 3, 6, 11$.

Recall that the cancellation laws state that if $ab = ac$ with $a \neq 0$ then $b = c$ and if $ba = bc$ with $b \neq 0$ then $a = c$.

Theorem 12.8 (Cancellation laws in rings)

The cancellation laws hold in a ring R if and only if R has no zero-divisors.

Proof.

Suppose first that R is a ring for which the cancellation laws hold, and suppose for contradiction that $ab = 0$ for some $a, b \in R$, $a, b \neq 0$. Then $ab = a0$, so $b = 0$, by left cancellation, a contradiction.
Suppose then that R is a ring with no zero divisors and let $ab = ac$ in R where $a \neq 0$. Then $0 = ab - ac = a(b - c)$ and since R has no zero divisors we have that $(b - c) = 0$, i.e. $b = c$. That is, the left cancellation law holds. A similar argument shows that the right cancellation law holds. □

Thus rings without zero-divisors are rings that behave nicely in some way. Commutativity and the existence of unity are also nice properties to work with. There are many useful examples of commutative rings with unity that have no zero divisors, for example the integers $\mathbb{Z}$. It is therefore useful to give a name to such an object:

Definition 12.9 (Integral Domain)

An *integral domain* D is a commutative ring with unity containing no zero divisors.

Example 12.10

(a) The ring of real numbers $\mathbb{R}$ is an integral domain.

(b) $\mathbb{Z}_p$, for p prime is an integral domain (using Corollary 12.6).

(c) Let R, S be commutative rings with unity. Then the direct product $R \times S$ is a commutative ring with unity, but if $r \neq 0, s \neq 0$ then $(r, 0)(0, s) = (0, 0)$ shows the existence of zero divisors, so $R \times S$ is not an integral domain.

(d) $\mathbb{Z}_2$ is an integral domain (it is commutative, the unity is 1, and it has no zero divisors), but the ring of 2×2 matrices with entries from $\mathbb{Z}_2$ is not an integral domain since

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

shows the existence of zero-divisors.

Example (Example 12.10 continued)

(e) We show that $\mathbb{C}$ is an integral domain. It is a commutative ring with unity 1. Suppose that $a+bi, c+di$ are non-zero elements of $\mathbb{C}$ such that $(a+bi)(c+di)=0$. Then $(ac-bd)+(bc+ad)i=0$, i.e. $ac-bd=0$ and $bc+ad=0$. If $c\neq 0$ then $a=bd/c$ so $bc+bd^2/c=0$, i.e. $b(c^2+d^2)=0$ so (using part (a)) $b=0$, and hence $a\neq 0$ (since $a+bi\neq 0$) and $ad=0$ so $d=0$, and hence ($ac-bd=0$ implies that) $a=0$, a contradiction. If $c=0$ then $bd=0$ and $ad=0$ and $d\neq 0$ so $b=0$, contradicting $a+bi\neq 0$. Thus $\mathbb{C}$ is an integral domain.

Lemma 12.11

Let R be an integral domain, and let S be a subring of R that contains a unity. Then S is an integral domain.

Proof.

Since R is a commutative ring, so is S. Since R has no zero divisors, S also has no zero divisors (any zero divisors of S would be zero divisors of R). Since S has unity, we therefore have that S is an integral domain. □

Example 12.12

$\mathbb{Z}$ is a subring of the integral domain $\mathbb{R}$ containing unity, so is an integral domain.

Theorem 12.13 (Fields are integral domains)

Every field F is an integral domain.

Proof.

Exercise 12.19. □

Recall that $\mathbb{Z}$ is an (infinite) integral domain, but is not a field. We now show that for finite rings, the concepts of field and integral domain are the same.

Theorem 12.14 (Finite integral domains are fields)

Every finite integral domain is a field.

Proof.

Let D be a finite integral domain. Then D is a commutative ring with unity, 1, say. We must show that every non-zero element is a unit. That is, for all $a \in D$ with $a \neq 0$, there is some $b \in D$ such that $ab = 1$.
Now since D is finite it has finitely many distinct elements: 0 and

$$1, a_1, \ldots, a_n \qquad (*)$$

(say). Now consider

$$a1, aa_1, \ldots, aa_n. \qquad (**)$$

If $aa_i = aa_j$ for some $a_i \neq a_j$ then by left cancellation (see Theorem 12.8) we have $a_i = a_j$, a contradiction. Thus all of these elements (**) are distinct. Since D is an integral domain, they are all non-zero. Thus, since the two lists have the same number of elements, the elements listed at (*) are the same as the elements listed at (**), in some order. In particular, we have $1 = a1$ or $1 = aa_i$ for some i. Either way, we have that a is a unit. □

Corollary 12.15

If p is prime then $\mathbb{Z}_p$ is a field.

Proof.

$\mathbb{Z}_p$ is an integral domain (by Example 12.10(b)), so the result follows from Theorem 12.14. □

Exercise 12.16

Find all solutions of the equation $x^3 - 2x^2 - 3x = 0$ in $\mathbb{Z}_{12}$.

Exercise 12.17

Find all solutions of the equation $x^2 + 2x + 2 = 0$ in $\mathbb{Z}_6$.

Exercise 12.18

Let R, S be integral domains. Find all zero divisors in the direct product $R \times S$.

Exercise 12.19

Show that every field is an integral domain.

Exercise 12.20

Recall the definition of idempotent from Exercise 10.20. Let D be an integral domain. Show that D contains exactly two idempotents.